



Customer Churn Prediction in Banking: A Comparative Machine Learning Approach

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EXECUTIVE SUMMARY

There is a growing concern over increasing customer churn in the banking industry, caused by increasing competition in the marketplace, changing consumer needs, and advancements in digital banking technology. Increased customer turnover creates a risk to the profitability of a bank, resulting from lost recurring revenues, increased cost of acquiring new customers, and decreased long term value of those customers. Therefore banks have been employing predictive analytics and machine learning to identify at risk consumers who will end their relationship with a financial institution. This study used three machine learning algorithms to evaluate whether they could effectively predict which consumers would leave a bank and assist in developing effective strategies for retaining customers.

The overall goal of this study was to determine if machine learning algorithms could predict customer churn among banking consumers and assess the ability of three different machine learning methods (Logistic Regression, Random Forest and Gradient Boosting) to accurately forecast customer churn and provide useful information for the development of strategies for retaining customers. A large publically available dataset on consumer banking relationships (Kaggle) was analyzed using a quantitative predictive model.

This study followed a structured methodological process, including data cleaning/preparation, exploratory data analysis (EDA), predictive modeling, and assessment of predictive performance. Data preparation activities included identifying/repairing missing values, evaluating outliers, converting categorical variables into numerical format, normalizing/scaling features, dividing data into training and test datasets, and correcting for moderate levels of class imbalance. As part of EDA, this study sought to gain insight into why consumers left their banking relationship, and identify significant factors that influence customer churn.

Results indicated a significant number of consumers were at risk of churning (churn rate = 20.37%), thus creating a need to develop a plan to retain existing banking consumers. Results from both the EDA and predictive modeling phases of this study indicate several factors contribute to consumer churn including but limited to consumer age, geographic location, inactivity, and account balance. Specifically, older consumers, consumers who do not use their accounts regularly ("inactive"), and consumers based in Germany had higher rates of churn than younger consumers and those residing outside of Germany. Additionally, results from this study demonstrate that consumers holding larger balances were more likely to close their account and cease doing business with the bank, implying that even large depositors are susceptible to "attrition".

Of the three machine learning techniques assessed in this study (Gradient Boosting, Random Forest and Logistic Regression), Gradient Boosting achieved the highest predictive performance with an accuracy level of 87.2% and a ROC-AUC score of 0.871. Random Forest generated highly accurate classifications as well. However, although Logistic Regression produced lower levels of accuracy than both Random Forest and Gradient Boosting, it did produce more interpretable results than both competitors.

Overall, this study demonstrates how machine learning techniques can enhance a bank's ability to proactively retain customers, improve its customer relationship management capabilities and create long term profitability opportunities.

Ultimately this study provides practitioners operating in the banking industry with valuable insights regarding how to utilize machine learning techniques to identify potential high-risk customers who are most likely to stop conducting business with them and implement focused retention strategies designed to minimize future customer churn.

CHAPTER 1 — INTRODUCTION

1.0 Chapter Introduction

The importance of retaining customers is rapidly becoming a major strategic priority for banks because of increased competitive pressures, a growing number of fintech companies offering online banking services, and evolving customer needs. With an increase in fintech competitors and new online banking platforms, switching from one service provider to another has become much simpler. Therefore, customer turnover or "churn" has become a significant concern for banks that wish to create long term profitable customer relationships and sustainable revenue.

1.1 Background of the Study

Customer Churn is defined as the Process in which Customers terminate their Relationship or Severely Reduce Their Association with Service Providers. In Banking Customer Churn can be described by means of Customers Closing Accounts, Transferring Financial Transactions to Competing Institutions or simply being inactive over Time. Due to the Highly Competitive Nature of the Banking Sector, Retention of Existing Customers has been growing in Importance for Banks who wish to maintain their Profitability and build Stronger Long Term Relationships with Customers.

In most cases retaining your current base of customers is viewed as more cost effective than Acquiring New Ones. Acquiring New Customers can be a very costly exercise including Substantial Marketing Expenditure, On-Boarding Costs and Relationship Building Activities, while Retaining Your Current Base of Customers typically Requires Less Financial Resources (Grönroos, 1994). Furthermore, Existing Customers can Generate Higher Levels of Profits to the Bank through Repeated Transactions, Increased Trust and Broader Usage of Banking Products and Services. This Concept is Closely Related to the Calculation of a Customer's Lifetime Value (CLV) which Refers to the Total Amount of Money Expected to be Generated by a Customer Over their Entire Life Cycle (Gupta & Lehman, 2003).

The Growing Digitalisation of Banking Services has altered Customer Expectations concerning Accessibility, Quality of Service and Convenience. Traditional Banking Firms are now Under Pressure from Digital Banks and FinTech Organisations Who Offer Faster, More Personalised and Technology Driven Services. Consequently Customer Loyalty is Becoming Increasingly Fragile, thereby necessitating Banks to Pro-Actively Identify Customers at Risk of Leaving. The Ability to Forecast Churn Before Customer Departure Can Allow Organisations to Develop Targeted Retention Strategies, Improve Service Quality and Minimise Revenue Losses.

1.2 Problem Statement

Customer churn forecasting is an important issue for banks because of the many and complex factors that are associated with customers' behavior, and therefore the difficulty in determining the cause of customers' decision to switch. Traditionally, Statistical Techniques (Logistic Regression) were applied to analyze customer defection and to determine which variables are most important in explaining it. These methods are well known for their ability to explain results clearly and transparently. However, these traditional Statistical Techniques may be limited in their ability to detect more complex or nonlinear relationships contained within very large sets of customer data (Hosmer et al., 2013).

Advances in machine learning recently provided more advanced analytic methods than previously available, enabling detection of underlying behavioral patterns and improved predictions of customers' likelihood of churn. Machine learning algorithms can examine large volumes of historic customer data to categorize customers based on how likely they are to churn (Bishop, 2006). While there has been increasing use of predictive analytics for predicting customers' likelihood of churn, few studies have compared the relative effectiveness of various machine learning models when used for predicting customer churn in banking environments. The purpose of this study was to address the gap identified by evaluating machine learning methods for predicting customer churn, and to identify effective strategies for supporting customer retention.

1.3 Research Aim and Objectives

The main goal of this study is to predict customer churning in the banking industry by using machine learning approaches, as well as comparing the predictive capabilities of the examined classification models.

In order to reach the stated goal, the study will address the following goals:

1. Identify the most important factors that contribute to customer churning in the banking sector.
2. Perform an Exploratory Data Analysis (EDA) on customer behavior, with a focus on identifying tendencies or patterns.
3. Implement logistic regression, random forest, and gradient boosting algorithms, and compare their performance in predicting customer churning.
4. Evaluate the performance of each of the aforementioned models by means of common classification metrics such as accuracy, precision, recall, f1-score, and roc-auc.
5. Develop business-oriented recommendations concerning customer retention.

1.4 Research Questions

This study aims to answer the following three research questions (RQ) that will allow us to explore these aspects of bank customer relationships.

1. What are the most significant factors of influence on customer churn in the banking sector?
2. Which Machine Learning Model has the highest accuracy with regard to predicting bank customer churn?
3. How can Predictive Analytics be used by banks to develop a Customer Retention Strategy that is Proactive rather than Reactive?

1.5 Scope of the Study

This research has an emphasis on bank customers' churn forecasting by use of public data set for banking (available at [kaggle.com](https://www.kaggle.com)) . Data set contains 10,000 records about bank customers with many attributes that can be used to explain bank customers' churn behavior. The analysis will be done by a variety of Supervised Machine Learning Methods including Logistical Regression , Random Forests and Gradient Boosting .

The primary goals of this research are to analyze predictive performance in the context of the features presented by the data set and to provide business users some interpretations of the results based on those features. Due to constraints of time and the scope of the project, no deeper analyses have been conducted; therefore, these conclusions cannot reflect all characteristics of banking population.

1.6 Significance of the Study

This study will contribute to both scholarly knowledge as well as practical applications for managers. In addition to expanding existing literature on customer churn prediction (Verbeke et al., 2012) using advanced machine learning techniques versus traditional statistical methods within a banking setting; this study can also provide banking institutions with data to identify their most at risk customers and implement strategic retention plans that target those at risk of leaving the institution and improve overall customer loyalty.

1.7 Structure of the Dissertation

This dissertation has been arranged over a period of seven chapters. The first chapter is an introduction to this study that outlines its purpose (research objective) and range (scope). The second chapter provides a review of existing literature on machine learning and customer churn. The third chapter details both the methods used for conducting the analysis as well as how it was carried out. In addition to providing descriptive information about the dataset in Chapter 4, Chapter 4 also carries out exploratory data analysis. In Chapter 5, the process of building models is described, along with the results from those models. Chapters 6 and 7 interpret the business implications of these models as well as the implications they have for management; finally, Chapter 7 is a summary of the conclusions drawn in this study, including its limitations and possible avenues for additional studies.

1.8 Chapter Summary

This chapter provided the background for the study; that is, it provided an overview of the research question (problem), the objectives of the study and the reasons why this study was conducted. The next chapter provides a literature survey of the areas of customer churn and predictive analytics in banking.

CHAPTER 2 —LITERATURE REVIEW

Introduction

The Literature Review Chapter is a critical evaluation of all of the theoretical knowledge and empirical evidence that is available on the subject matter (customer churn prediction) in relation to customer churn prediction in the Banking Sector. Banks need to retain their customers in order to remain profitable, and provide themselves with an opportunity to establish long term relationships with customers, especially since competition among banks for customers' deposits has increased significantly over recent years. Customer churn is an issue because it results in lost revenue, higher costs associated with acquiring new customers, and a lower customer life-time-value. For this reason, predicting which customers will churn has become an extremely important field of study for both academics and professional practitioners.

Statistical analysis has traditionally been used to assess how customers behave. However, traditional statistical analysis can be ineffective at capturing the complexity of customer behavior. Because of its ability to analyze large amounts of data and identify complex customer patterns and relationships, Machine Learning has increasingly been employed to enhance the accuracy of churn predictions and to assist in making informed business decisions. This Chapter includes a comprehensive literature review of research pertaining to customer churn in the banking industry, customer retention theory, machine learning methodologies applicable to churn prediction, and previous empirical research studies. Finally, the Chapter describes the gap in the existing research that this study addresses.

2.1 Customer Churn in Bank

2.1.1 Define Customer Churn

The term "Customer Churn" in Banking refers to when customers stop using the Bank or decrease their use of Bank Services. This could occur for several reasons such as, Closing Accounts, Becoming Inactive, Switching to Competitors that are providing better services. Based on Continued Engagement with a Bank's products/services customers will fall into two categories: Retained Customers/Churned Customers. Voluntary Churn is typically caused from being dissatisfied or finding an alternative service/products that is perceived to be better. The other type of Churn is called Involuntary Churn and this type occurs because of Fraudulent Activity or Regulatory Issues. High levels of Churn therefore can cause lower Profitability and/or Competitive Advantage along with Lowered Customer Lifetime Value.

2.1.2 Why Customers Leave Banks

Factors influencing bank customer churn include various behavioural, financial and demographic elements. For example, one of the main reasons for customers leaving their bank is due to a lack of good service quality, i.e. poor customer support and a slow response time when problems occur. In addition to these issues, an increasing number of customers are expecting faster, better and more responsive services from their banks, and if they do not receive this then it can lead to dissatisfaction with the bank which may lead to switching to a competitor (Zeithaml et al. , 2000). A high fee or additional charge from their bank will often result in customers looking for an alternative that offers similar products but at a lower cost. The emergence of digital banking and fintechs have opened up new opportunities for customers who want simple access via mobile banking to personalised digital experiences and fast financial solutions (Verhoef et al, 2021). Poor personalisation of the experience by the bank and ineffective communication may further damage customer satisfaction. Demographic factors including the age, income and geographic location of the customer will affect how the customer perceives and reacts to different options available for them and therefore affect long term customer retention (Verbeke et al., 2011) .

2.1.3 Why Churn Matters to Banks

Customer attrition is a major issue in banking due to lost customers result in less money being made on products & services. The more you lose customers (churn), the higher your cost of acquisition will be to get new customers. Losses in terms of cost aren't all that customer loss has, it can also have an impact on your reputation with customers, their level of trust in you, and where you compete with others.

2.2 Customer Retention Theory

Customer Retention Theory - Understanding Why Banks Invest in Customer Retention

2.2.1 Acquisition vs Retention Cost

In many highly competitive markets like banking, acquiring new customers will be more expensive than keeping existing customers, a major reason for this is that it requires larger marketing and advertising expenditures, and is also much more expensive to onboard (Kotler & Keller, 2015). As an example, maintaining customers that have used your services before would be easier and less expensive than attracting new customers, in most cases, customers retain familiarity with the organization's brand and offerings. Relationship marketing theory suggests that keeping customers long term can help to increase their trust and loyalty toward an organization, which results in increased engagement from those customers (Grönroos, 1994), resulting in increased potential for future revenue through repeat sales, and purchasing of additional products and/or services. Therefore, businesses are increasingly focusing on

developing customer retention strategies to lower customer churn rates, improve their customer base, and grow their organizations sustainably.

2.2.2 Customer Lifetime Value (CLV)

The value that can be generated by a bank from a customer over time is known as the customer's "lifetime value" or CLV. This shows how much of a bank's profit will come from a long-term view of its customer base instead of just looking at one transaction at a time. Customers who generate more lifetime value are likely to provide a large amount of money to a bank through saving, lending, investing and other services. As such if these customers leave through churn it could have a negative effect on profitability and overall business performance. For this reason, churn prediction models enable banks to identify which customers are most likely to churn and take action to retain them before they do.

2.2.3 Role of Data Analytics in Retention

Customer Retention Using Analytics is now a key area for business as it helps organisations find out which of their customers will be leaving. Ngai, Xiu & Chau (2009) note that predictive analysis can help companies understand customer behaviour and see when they have started to develop 'churn' so the company can intervene early. The ability to target customers through specific personalisation strategies such as, loyalty schemes or direct marketing/retention campaigns increases the likelihood of customer interaction. Customer relationships are also enhanced with the aid of more informed decisions made based on customer insight from analytics.

2.3 Machine Learning in Churn Prediction

2.3.1 What is Machine Learning

Machine learning is the area of artificial intelligence (ai) which allows systems to learn patterns in data and predict outputs based on that learning without being explicitly programmed (Bishop, 2006). Supervised learning techniques are commonly used for predictive analytics, these models are trained using labeled historical data to make predictions about future outcomes. Customer churn prediction is typically considered as a binary classification problem where customers are classified into two categories - either churned or retained. The use of machine learning techniques allows organizations to identify hidden customer patterns and improve decision making through more accurate prediction models.

2.3.2 Logistic Regression

Logistic regression is a type of classification model that can be employed to forecast binary outcomes (e.g. will customers churn or continue to bank) by estimating the probability of occurrence of an event on the basis of the relationship among independent variable(s), a set of explanatory variables, and a single dependent variable or target outcome (Hosmer et al., 2013). As one of the most popular statistical methods for predicting churn, logistic regression has many advantages. Its use is quite common due to the simplicity and high interpretability of models. Additionally, since model coefficients allow identification of which customer attributes have positive and negative influences upon churn probability, they are also very easy to explain. Nonetheless, although logistic regression is widely accepted, it may not easily be able to describe the often very complex, nonlinear relationships that exist in large datasets related to customers. Thus, in this research, logistic regression was selected as the base case for all performance comparisons.

2.3.3 Random Forest

An advantage of random forest in predicting churn behavior for customers is the strength of the model to interpret high levels of complexity in both linear and non-linear patterns present in the customer's dataset. As well, one of the greatest advantages of this model is it has higher accuracy when compared to other simple models in use today. A disadvantage of the model however, is that the level of interpretation is lower compared to simpler models such as logistic regression. In addition to providing stronger predictive performance, the Random Forest model also allows the researcher to determine which features are most important based on how they relate to churn behavior.

2.3.4 Gradient Boosting

Gradient Boosting uses an enhanced method for creating ensembles of predictive models based on sequential decisions. Each of the additional decision trees is designed to reduce error left over by previously built decision trees. Therefore, unlike the independent building of trees used in Random Forest, Gradient Boosting creates models that build on top of each other with iterative corrections for past mistakes. One advantage of using this methodology is its ability to create highly accurate predictive models that can represent non-linear and complex customer behavior characteristics. This is one reason why Gradient Boosting is commonly used as part of customer churn predictive modeling processes. Additionally, Gradient Boosting is considered computationally intensive compared to other methodologies, and can be more difficult to interpret as it does not provide the same level of clarity into how specific input variables are influencing the model's predicted outcomes.

2.3.5 Why These Models Were Selected

The three Machine Learning models have been picked based on a balance of their capability to provide explanation, robustness and good prediction capabilities. Logistic regression has been chosen as it is very easy to understand and interpret how customer behavior will affect churn by the variables that make up each customer's characteristics (Hosmer et al., 2013). A random forest model has been chosen for its robust nature and to be able to identify the relationship between multiple variables and how they interact with one another. Finally, gradient boosting has been used for its high predictive accuracy and ability to identify hidden trends in the customer dataset (Friedman, 2001, Breiman, 2001).

2.4 Previous Studies

Theme 1: Traditional Statistical Models

Traditional Statistical Models such as Logistic Regression have become very popular for predicting customer churn because they are simple to use and easy to explain. They work well with research in this area because they allow researchers to easily see how each independent variable affects a customers' likelihood to churn while providing good enough predictive results. (Hosmer, Lemeshow & Sturdivant, 2013). In fact, according to (Verbeke et al. 2011), even though interpretable statistical models can be used to predict customer churn by giving business owners an easier way to see what factors contribute to a customers' tendency to churn, they still hold value in today's data-driven world.

Theme 2: Tree-Based Models

Random forest as well as decision trees have grown in popularity for use in predicting customer churn because they are able to determine the non-linear and complex relationship among all possible characteristics of a customer. Traditional statistical models do not perform as well with respect to identifying interaction among multiple variable types (Hadden et al., 2007), which is why random forests and other tree based models are beginning to be preferred over logistic regression as a method of identifying customer behavior patterns that result in churn (Amin et al., 2019). Additionally, random forests are capable of producing high levels of accurate predictions about customer churn, and provide feature importance values that help explain what characteristics or behaviors are most likely to lead to a customer churning.

Theme 3: Boosting Models

Gradient Boosting Models and XGBoost Models are now being used a lot in customer churn modelling as they can classify customers very well. The way Gradient Boosting and XGBoost Models work is based on sequential learning. This means that after every model built from the data set is run against the training data, another model will be created and run

against the same data set with the intention of correcting any mistakes the previous models made (Friedman, 2001). Previous research has shown that using a boosting technique produces better predictive results compared to traditional statistical analysis or decision trees when predicting customer churn, especially in large customer databases (Chen & Guestrin, 2016).

Theme 4: Important Churn Predictors

Past studies have shown that a variety of key indicators for customer churning include: Age, Status as Active or Inactive, Account Balance, Geographic Location, and Income Level. Researchers have indicated that inactive customers, older customers, and less engaged customers are generally most likely to terminate their relationship at a bank (Keramati et al, 2014, Verbeke et al, 2011). Geographical and Financial factors can be a source of difference in switching behavior due to regional customer preference patterns and local economic factors. These same variables were found to be significant in this investigation as well, specifically: Age, Status as Active/Inactive, Account Balance, and Geography.

2.5 Research Gap

Although previous research has substantially contributed to the development of customer churn predictions, there are still substantial shortcomings to be addressed by future research. The majority of studies have concentrated their efforts to improve predictive power. As a result of this effort, studies generally concentrate on achieving higher levels of model performance, but give little consideration to either model interpretability or model explainability (Verbeke et al, 2011). Therefore, it can create problems for banks who require transparent decision-making processes, so that they may understand why a customer is likely to churn. In addition to the fact that few studies have provided a direct comparison of multiple machine learning techniques (such as Logistic Regression, Random Forest and Gradient Boosting) in relation to predicting bank customer churn. Finally, most studies do not emphasize developing practical business insights and retention strategies based upon the results derived from model outputs.

Consequently, this study will seek to compare both the predictive performance and interpretability of multiple machine learning techniques used to predict customer churn in banking, and ultimately derive business insights from the output(s) generated from each technique.

CHAPTER 3 — METHODOLOGY

3.0 Methodology Introduction

The purpose of this Chapter is to outline how the Research was Conducted for Developing and Evaluating Customer Churn Prediction Models in Banking Industry. Methodology can be defined as an orderly procedure adopted by Researchers for collecting, analysing and interpreting Data that ensure Reliability and Validity of the Findings from the Research (Saunders et al, 2009). The Study uses Quantitative Research Paradigm based on Predictive Analytics and Machine Learning Techniques to Analyze the Customer's Churn Behavior. The Steps taken during the Research Process were Collection of Datasets, Cleaning and Preprocessing the Datasets, Exploratory Data Analysis (EDA), Development of Model, Evaluation of Performance of Developed Model, Business Interpretation of Results. Additionally, the Current Chapter explains Research Design, Description of Dataset, Methods Used for Preprocessing Datasets, Exploratory Analysis, Selection of Model, Metrics Used for Evaluating Performance of Machine Learning Model in Predicting Customer Churn (Creswell, 2003).

3.1 Research Design

This predictive analytics research examines how banks can use their customer data for predicting when customers will leave or "churn." Because this research is based on numeric data from bank customer accounts and uses statistical methods as well as machine learning to discover trends related to bank account churn and produce predictions regarding churn (Creswell, 2003), a quantitative methodological approach was chosen. Furthermore, by using predictive analytics, organizations may recognize which customers have a higher likelihood of churning in the near future, understand behavioral characteristics of those customers who do tend to churn, and assist in making proactive business decisions that enhance customer retention.

In addition, this research will utilize supervised machine learning classification to classify the type of relationship each customer has with the bank. As historical labeled data were available, and since the target variable in this study is also known, the customer's relationship with the bank will be classified as either "retained" or "exited". Because there are only two possible classifications (retained or exited) for customer status, this study is categorized as a binary classification problem (Bishop, 2006). The methodology chosen for this study is most suitable because it allows researchers to predict when customers will churn, permits comparisons of different machine learning algorithms, and provides researchers with a basis for providing useful and practical interpretations of results that relate to business decision-making.

3.2 Dataset Description

This dataset was accessed from the Kaggle Bank Customer Churn Dataset that has customer-level bank data for predicting customer attrition. This dataset contains customer records totaling 10,000, enough for a reasonable amount of data to support predictive modeling and enhance the reliability and robustness of model performance. A greater number of observations will improve the consistency of the results by lowering the risk of model overfitting and allow for better generalization of model performance to banking churn predictions.

There are eleven predictor variables and one target variable, “Exited,” that defines customer churn. For this research, the “Exited” value is 0 when the customer remains with the institution and 1 if the customer churns. These predictor variables are: **CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, and EstimatedSalary. All of these variables have the potential to affect how customers behave regarding their loyalty/retention to an institution.

There are various types of variables included in the dataset. Examples of numerical variables are, Age, Balance, EstimatedSalary, and CreditScore. Each type provides specific types of quantitative or measurable customer financial/demographic information. Categorical variables such as Gender and Geography identify group characteristics of customers. Binary variables such as HasCrCard, IsActiveMember, and Exited can be identified as either a "yes" or "no" based on whether a characteristic exists.

In addition to other considerations there were also some class imbalances present in the dataset. Approximately 80% of the customers remained with the institution and approximately 20% left the institution. A stratified train-test split was conducted to adjust the proportionality of the classes at the time of data preparation.

3.3 Data Preprocessing

3.3.1 Missing Values

The dataset was checked for missing values in order to assess the quality and reliability of the data before modeling. The analysis found that there were no missing values. Missing data can add an element of bias to the model, decrease prediction accuracy and generally have negative effects on how well a machine learning model will perform (Han, Kamber and Pei, 2011)

3.3.2 Outlier Detection

Boxplots were used to detect outliers for numeric variables CreditScore and Age and balance and estimatedsalary. A number of outliers were found notably in Age and credit score. The outlying data points were however kept as they could reflect how customers behave and there is natural variation within banking data, furthermore tree based models have been shown to be relatively robust to the influence of extreme or unusual values (Aggarwal, 2017).

3.3.3 Encoding Categorical Variables

Since machine learning model inputs need to be numeric, categorical variables are converted into a numeric representation prior to modeling. Gender is label encoded with values of 0

for Female and 1 for Male. The geographic data from the survey was one hot encoded which produced new columns such as Geography Germany and Geography Spain and France was left as the base category to prevent the dummy variable trap and potential multicollinearity problems that can arise when all categories are included in a model (James et al, 2021).

3.3.4 Feature Scaling

Feature scaling was used with StandardScaler to enhance the model's performance, especially for logistic regression, because it is affected by large variations in the scale of variables. StandardScaler, scaled the numerical features so that they were all centred on 0 mean and had a standard deviation of 1. This ensured that each feature contributed equally to the prediction. Random Forest and Gradient Boosting are generally less impacted by scaling than logistic regression (Géron, 2022).

3.3.5 Train-Test Split

The data set was divided into 80 percent training data (8,000 observations) and 20 percent testing data (2,000 observations) to allow for an accurate model performance assessment. A random state value of 42 was used in order to ensure results were reproducible. Stratification on the churn class was also performed using stratify = y, which helped to preserve approximately 79.6 percent retained and 20.4 percent churned customers in the training data set and 79.6 percent retained and 20.3 percent churned customers in the test data set (James et al, 2013).

3.3.6 Class Imbalance

The data set had an approximate of 20 % churned customers that made for a moderate level of class imbalance. A stratified train/test split was used to address this issue by preserving the class distribution in both sets. In addition to accuracy as a measure of performance, Precision, Recall, F1-Score, and ROC-AUC were also used to evaluate the model's performance (James et al, 2021).

3.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) provided insight into how the customer behaved in order to help discover patterns associated with churn, in addition to determining the level of interaction among the independent variable(s), and, most importantly, which features are significant indicators for predicting customer turnover. (Han et al. 2022) explain that "exploratory data analysis is the first step in all statistical analysis" because it provides an opportunity to see how the underlying structure of your dataset behaves prior to developing

models. The analyses I completed allowed me to study and gain knowledge about the behavior of churned customers.

Churn Distribution analysis determined that approximately 20.37 percent of customers churned, which indicates a moderate degree of class imbalance. Gender analysis shows that females exhibited higher churn rates than did males. Region-wise, the geographic area with the highest churn rate was Germany. Age-wise, customers who were older were more likely to churn. Inactivity Status analysis confirmed that inactive customers displayed significantly more churn behavior. Additionally, correlation analysis illustrated that age is the strongest positive predictor of churn. Conversely, the IsActiveMember feature demonstrates a negative relationship with the probability of churn. Collectively, EDA assisted in developing understanding regarding each feature as well as improving interpretation regarding the results from machine learning models.

3.5 Model Selection Justification

Logistic Regression

Logistic Regression was the first choice of the baseline model due to its relative simplicity and high level of interpretability. It is one of the best known Statistical Classification methods which uses predictor variables to estimate the likelihood of a customer churning. A key advantage of using logistic regression is that it produces easily understandable coefficients which can be interpreted clearly to understand how customer behaviours affect customer churn (Hosmer et al, 2013). One of the key disadvantages of using logistic regression is that it will find difficulty identifying non linear or complex relationships in customer behaviour data.

Random Forest

Random Forest has been selected because of its potential for providing good robustness & predictive performance in classification tasks. An example of Ensemble Learning Techniques, Random Forests are made up of many Decision Trees working together to achieve better predictive performances than could be achieved by individual Decision Trees alone (Breiman, 2001). Additionally, Random Forest's use of feature importance allows researchers to identify those predictor variables most likely to be associated with customer churn. Although Random Forests do provide good predictive performance and useful feature importance information, they are generally seen as being harder to interpret than other machine learning techniques because of their complexity.

Gradient Boosting

Gradient Boosting has been selected because of its strong predictive power & ability to improve predictive power through iterative learning. Similar to Random Forest, but unlike Random Forest, Gradient Boosting makes decision trees sequentially by adding new decision trees to try to correct the errors of the previous tree(s) (Friedman, 2001). Generally, gradient boosting will produce the highest predictive accuracy & perform better at classification tasks than random forest. However, the increased complexity of gradient boosting models also means they can potentially be less easy to explain and therefore less easy to interpret.

3.6 Evaluation Metrics

Accuracy

Accuracy provides an estimate of the ratio of correct classifications in relation to the total number of predictions that have been made by a given model, hence, it serves as an indicator of how well a model performs. Yet, using this one method for assessing how well a model performs with data where there is a significant difference between the two classes (majority/minority), will provide biased results toward the larger class. (James et al., 2021)

Precision

Precision evaluates the ratio of correct classifications of those customers who were actually churned out of the total number of customers who were predicted to churn. Thus, high precision is important to ensure that you do not incorrectly classify customers as being at risk of churning and therefore subject them to retention efforts unnecessarily. (Sokolova and Lapalme, 2009).

Recall

Recall evaluates the ability of a model to predict which customers in your database are likely to churn out. The importance of having a high recall in banking lies in the fact that if you fail to detect the customers who are likely to churn out from your customer base before they actually do, then you miss the opportunity to intervene early and retain these customers. (Fawcett, 2006).

F1 Score

The F1 score combines both precision and recall into one metric. It does so by averaging the two metrics together. Because it is based on the average of precision and recall, the F1 score is a good way to evaluate whether or not a model has achieved a good level of performance. When evaluating models that have imbalanced classes, the F1 score is also helpful since it balances performance against both positive and negative classes. (Sokolova and Lapalme, 2009).

ROC-AUC

The Receiver Operating Characteristic - Area under the Curve (ROC-AUC) is used to determine the degree to which a model can accurately separate customers who churn from those that do not. The ROC curve plots the true positive rate against the false positive rate at different thresholds. Models with higher ROC-AUC values perform better than those with lower values. Therefore, ROC-AUC allows for comparison of a model's quality along with its discriminative power. (Fawcett, 2006).

Cross-Validation

Cross-validation was employed to enhance the reliability of each model while reducing reliance upon a single train/test split. Consistency in the model was evaluated through mean ROC-AUCs, resulting in a mean ROC-AUC of 0.767 for logistic regression, 0.858 for random forest, 0.864 for gradient boosting, showing that these three models are superior to each other.

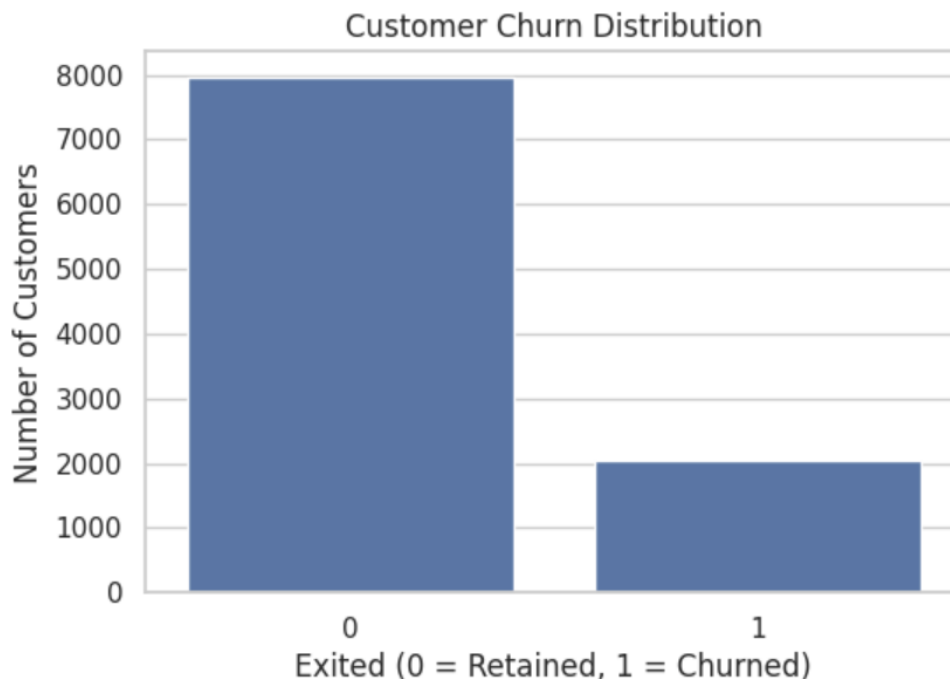
CHAPTER 4 — EXPLORATORY DATA ANALYSIS (EDA)

4.0 Introduction to EDA

Exploratory Data Analysis (EDA) was used in order to understand the shape and nature of the data before beginning the task of developing a predictive model. EDA is an important analytical methodology which can help find patterns in how customers behave, connections among variables, indicators for potential churn, irregularities or inconsistencies, and trends embedded in datasets (Han, Kamber & Pei, 2011). In addition, using EDA enhances the comprehension of features by researchers working on machine learning models, enables researchers to interpret their results from predictive modeling and ultimately improve the quality of predictions made by those models, all resulting in improved decision making as part of the model development process (Hair et al., 2019). This EDA contained analyses on general churn distributions, churn distributions by gender, geographic variation, age related

changes in churn rates, active/inactive customer statuses and relationship among variables so that researchers could have a greater understanding of what influences the loss of customers.

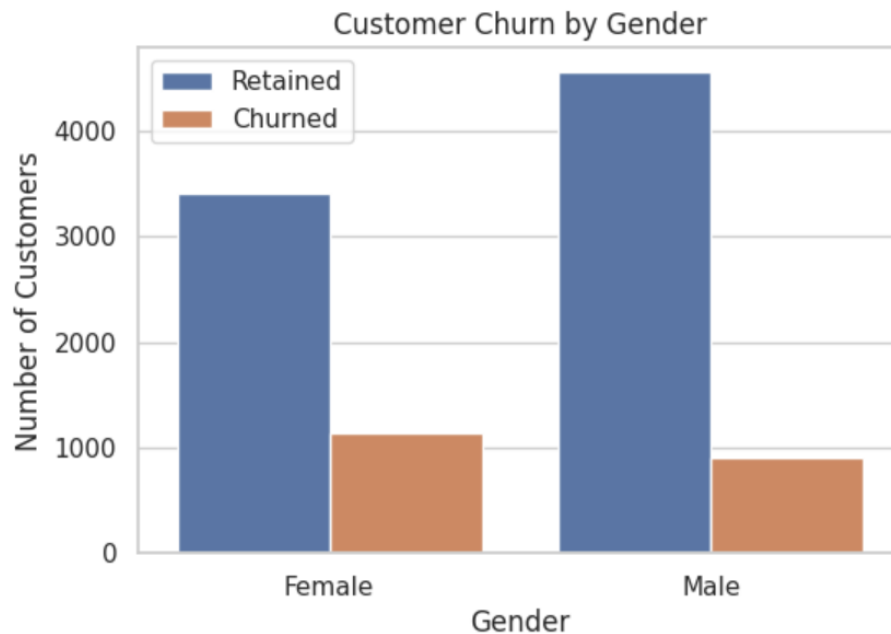
4.1 Churn Distribution



Customer churn was found to be distributed among customers, with 20.37 % exiting the bank, and 79.63% remaining. While there is an apparent class balance issue due to the higher number of retained customers, this higher rate of customer churn still represents a significant problem for banks. Approximately 20% or one out of five customers are leaving the bank, therefore, it is crucial for banks to understand why they lose these customers.

A loss of approximately 20% of its customers can result in a loss of a significant portion of the bank's profit, and a large amount of recurring revenue, and ultimately affect long term customer value (Kotler & Keller, 2015) Therefore, from a business standpoint, developing a predictive churn model will allow banks to identify their most at risk customers and develop proactive retention strategies. Additionally, the moderate degree of class imbalances within our data warranted us to evaluate model performance using multiple measures including Precision, Recall, F1-Score, and ROC-AUC to ensure we do not rely on Accuracy alone when evaluating our models' effectiveness.

4.2 Churn by Gender

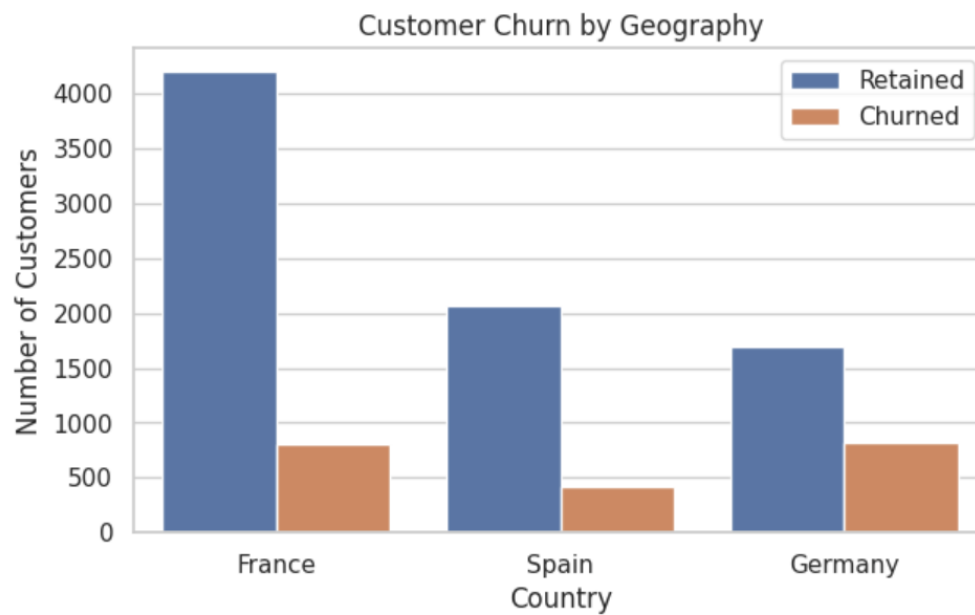


Gender based churn analysis has been used to assess if there is a difference in the likelihood of customer churn between male and female customers. Our results are consistent with those of previous studies indicating that females were more likely to experience churn than males, and thus, had a greater proportion of churning customers when compared to male customers, conversely, our results indicated that males experienced better retention rates than females. We would like to caution against over-interpretation of these results because it is possible that the gender of an individual does not independently impact their likelihood of experiencing churn.

There are several reasons for which differences in the propensity to churn could exist. Customer experiences and perceptions related to services provided, financial products preferred, expected service quality and level of engagement may all indirectly contribute to churn patterns among different genders. For example, the degree to which customer dissatisfaction with service quality or failure to meet financial needs impacts their decision to continue using a bank's services may differ depending on the type of banking services needed and how they are used.

As a result of these findings we believe that banks can profitably implement tailored marketing programs, customer engagement strategies and retention programs that target at risk customer segments. In doing so banks may increase customer satisfaction and enhance long term relationships. (Verbeke et al. 2011) also found similar evidence in their study on how demographic variables impacted the likelihood of churn behavior in predictive modeling applications.

4.3 Churn by Geography



Exited	0	1
Geography_Label		
France	83.845233	16.154767
Germany	67.556796	32.443204
Spain	83.326605	16.673395

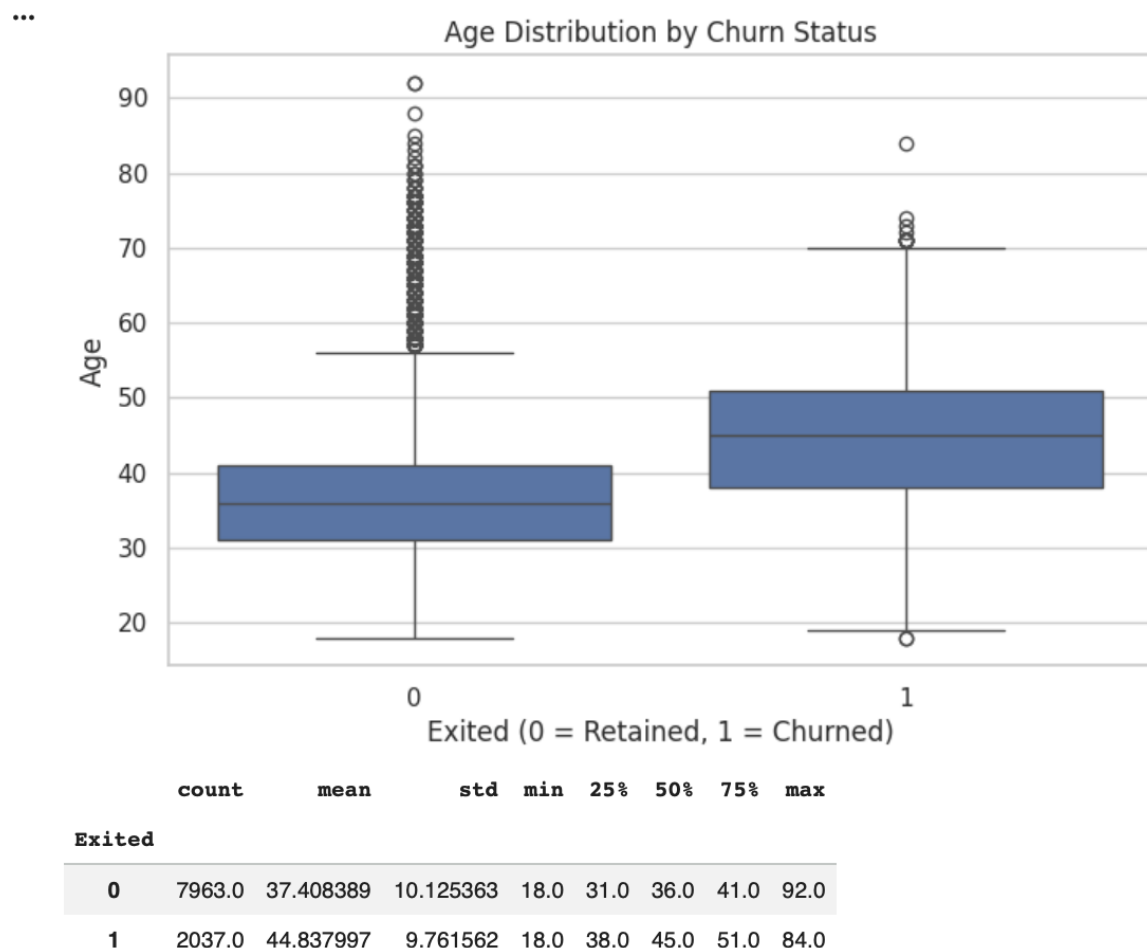
The geographic analysis performed for this study was conducted to assess whether there are variations in customer churn rates by region. From the data collected, it has been established that customer churn behavior varies significantly among countries. Specifically, France had an approximate retention rate of 83.85 percent and a corresponding churn rate of 16.15 percent. Additionally, Spain had a similar trend with an estimated retention rate of 83.33 percent and a corresponding churn rate of 16.67 percent. Conversely, Germany experienced the largest churn rate, with an estimated retention rate of 67.56 percent, and a corresponding churn rate of 32.44 percent, which represents nearly a double churn rate when compared to those of France and Spain.

Therefore, these results indicate that customer churn behavior is not consistent throughout geographic areas. There are potential reasons that could explain why customer churn behaviors vary based on geographic area including but limited to differences in customer expectations, competition levels, local banking preferences, and perceived service quality. When competitive banks provide improved digital services, pricing models, and/or customer experiences, customers in specific geographic locations can become much more likely to engage in switching behavior (Zeithaml et al. , 2000).

For businesses specifically, banks may find value in developing regionally focused retention efforts to address higher risk areas than those found in Germany. This includes engaging customers, understanding the elements contributing to customer dissatisfaction, and providing tailored services based on the expectations of customers within their respective

regions. Historically, researchers have demonstrated that geographic attributes affect customer switching behavior and resulting churn (Verbeke et al., 2011).

4.4 Age and Churn Analysis



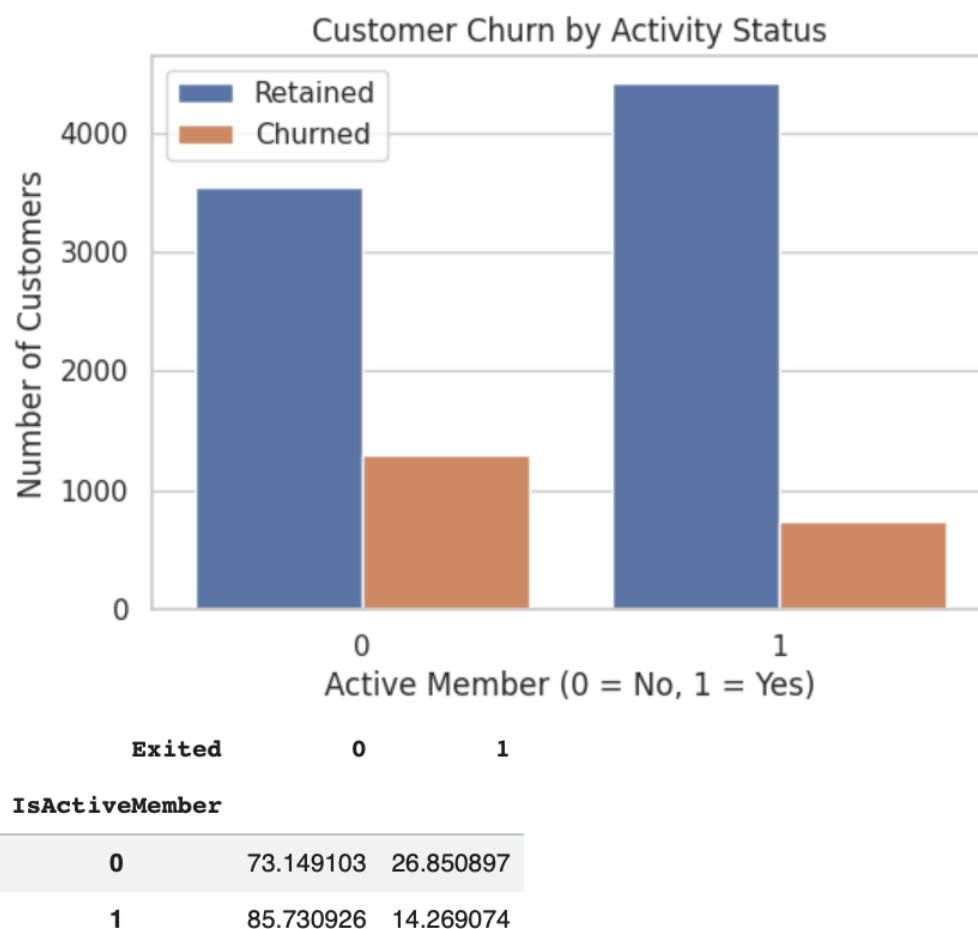
Age-based analysis was performed to examine the relationship between customer age and churn behaviour. The findings show clear differences in terms of mean and median values for retained and churned Customers. Retained Customers had a mean age of 37.4 years with a median value of 36 years, while churned Customers had a much older mean age of 44.8 years with a median value of 45 years. As such, churned Customers were substantially older than their retained counterparts indicating that there may be an interaction effect between age and customer attrition.

In addition to the observations made from the summary statistics, the boxplot also provides evidence of a large gap in the distribution of ages between churned and retained Customers. However, it is important to note that several outliers were found in this study, these outliers were kept in the data set to preserve realistic variability across all Customers. Possible explanations for why churned Customers are older than their retained counterparts include: (i) older Customers may have financial needs that change over time, (ii) older Customers may have higher service expectations or (iii) older Customers may seek better financial products/personalised banking solutions, which could result in them switching providers (verbeke et al, 2011).

From a business perspective, banks may benefit by segmenting Customers based on age, developing age-targeted financial products, strengthening engagement strategies for older customer groups. Furthermore, the importance of age as a key driver of churn can not be ignored as later it emerged as the strongest positive correlation with churn, as has been reported in previous studies (Gupta & Lehmann, 2003).

4.5 Activity Status and Churn

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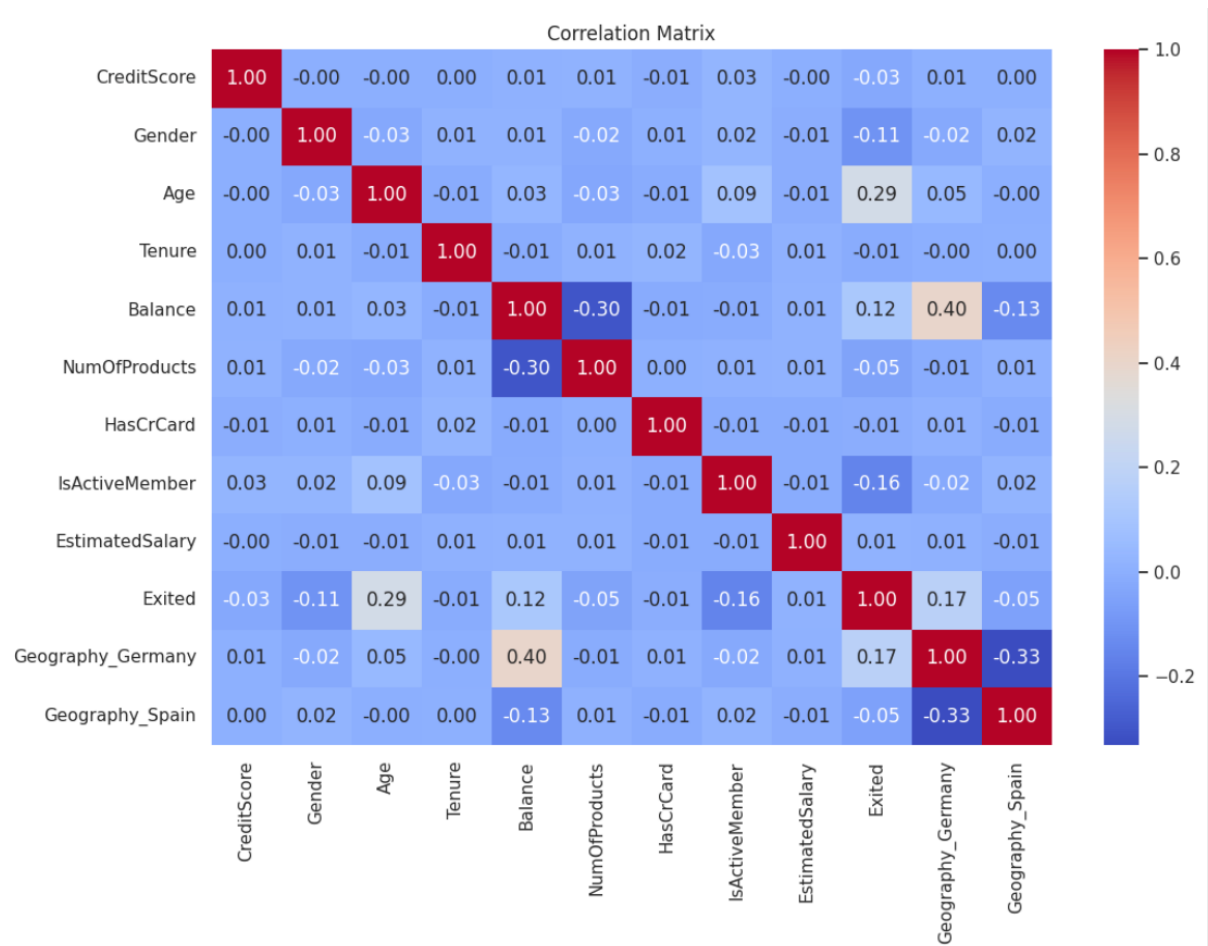


In this study we also identified the impact of being active or inactive on a customer's decision to churn. We found a difference between active and inactive customer groups in terms of churn behavior. A total of 73.15% of inactive customers were retained by the bank, whereas 26.85% churned. Active customers had significantly better retention as compared to inactive customers with 85.73% retained and only 14.27% churned. Thus, inactive customers have much higher churn rates than active customers.

Our research suggests that customer interaction is crucial for a positive outcome when it comes to retaining customers. Inactive customers are less likely to be loyal to the bank, they will use fewer banking services, therefore, they will be more likely to switch banks. Our results support relationship marketing theory which states that if the bank engages actively with their customers, the bank can increase trust and loyalty of their customers resulting in longer term retention (Grönroos, 1994).

The bank could potentially profit from identifying inactive customers earlier in order to offer them tailored financial products and initiate re-engagement initiatives to enhance their client base.

4.6 Correlation Analysis



Correlations have been computed to identify how predictor-variables are related to customer-churn. In this regard, the correlation heatmap illustrated numerous predictor-variables that are related to churn, however, nearly all relationships were relatively weak, as (James et al. 2021) reported to typically exist within customer-behaviour-datasets. The highest positive correlation found with churn was for Age (0.29), indicating that older customers are more likely to discontinue their banking services. A similar finding existed for Geography Germany (0.17), which indicated that German customers tend to churn at a greater rate. Balance (0.12) also exhibited a weak positive correlation, indicating that customers who maintain a larger account balance might be marginally more likely to churn from their banking services.

Conversely, a negative correlation existed for IsActiveMember (-0.16) and churn, which indicates that active members are more likely to remain customers. This supports prior exploratory analyses demonstrating the importance of member engagement to retain its customers.

As such, these data indicate that customer churn can be explained by various combinations of demographic and behavioral characteristics, rather than individual characteristic alone explain customer behavior. For business purposes, banks could use both demographic-based and behavior-based attributes and predictably utilize advanced analytical tools (predictive analytics) rather than using assumptions. Furthermore, the findings support the use of

machine learning algorithms that enable identification of complex associations between predictors and outcomes, thereby provide a solid base for developing predictive models in Chapter 7.

4.7 Chapter Summary

Exploratory Data Analysis (EDA), as described in this chapter was used to identify customer churn patterns. Important EDA results were that there is a medium level of churn, churn is higher for female than male customers, churn rates are greatest in Germany, and both age and time since last use of services demonstrate an increase in the probability of a customer leaving the bank. Age has been shown to be most strongly positively correlated with customer loss. As such, the data analysis work on customer features from this chapter improved our ability to interpret behavior of individual customers and informed our development of predictive models of customer loss for the next chapter (Han, Kamber and Pei, 2011).

CHAPTER 5 — MODEL IMPLEMENTATION & RESULTS

5.0 Introduction to Model Implementation

In this Chapter, a number of Machine Learning Models are implemented with the goal of predicting customer churn in banks. The goal is to evaluate and compare the predictive capabilities of three Supervised Learning Techniques: Logistic Regression, Random Forest, and Gradient Boosting. Additionally, due to the moderate imbalance of data within the training set, a variety of Evaluation Metrics will be used to provide an accurate representation of model performance (James et al., 2021) (Bishop, 2006), these include Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Each section will describe both the Implementation of the predictive Model and its Performance.

5.1 Logistic Regression Results

5.1.1 Model Overview

I chose to implement logistic regression for this problem because it has good interpretability in classification problems. Also, due to its ability to provide coefficients on each factor contributing to the customer's decision-making process (e.g., Hildebrandt et al., 2006,

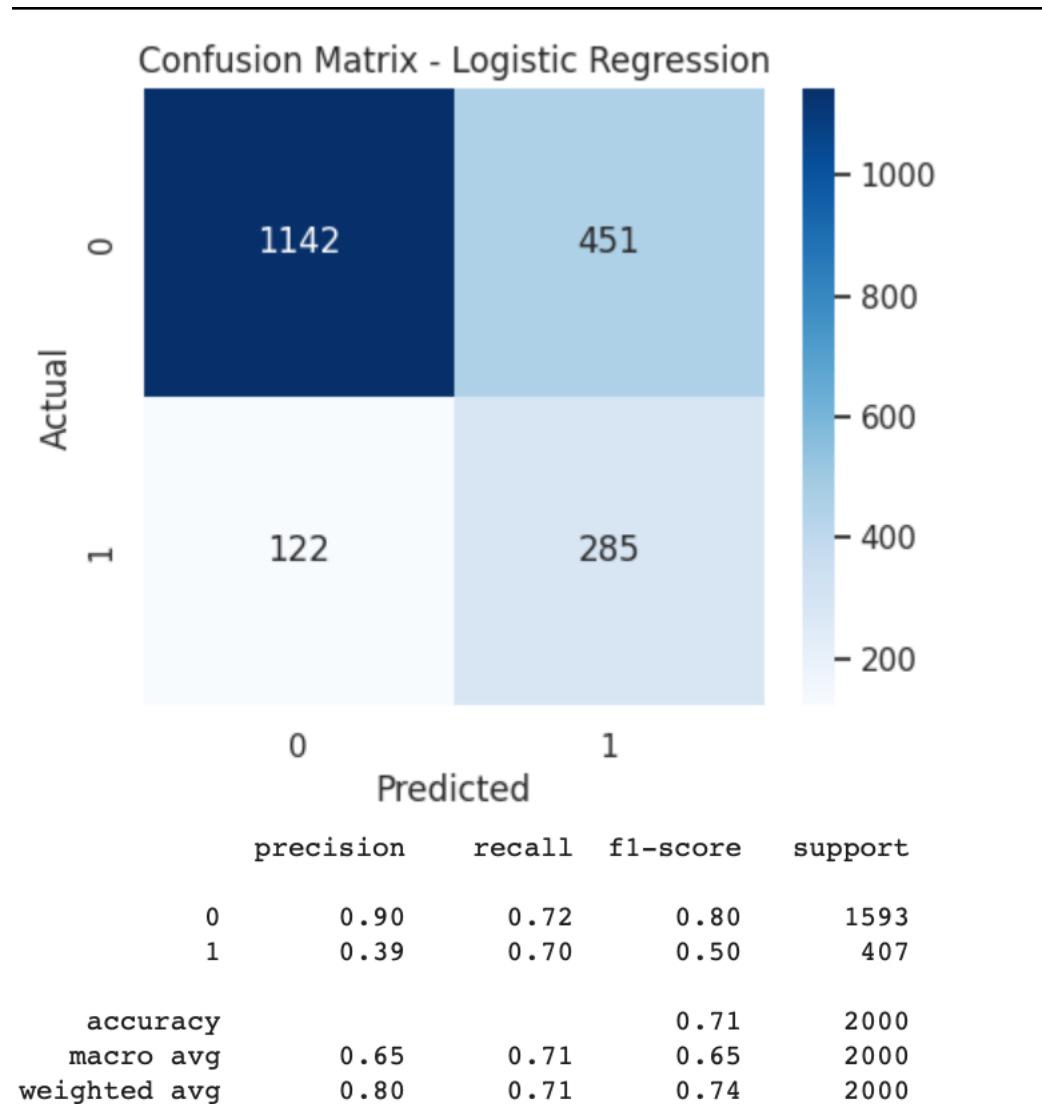
Hosmer, Lemeshow & Sturdivant, 2013) it has been widely used in customer churn prediction problems.

5.1.2 Model Performance

The model had an accuracy of 71.4%, a precision of 38.7%, a recall of 70.0%, F1-score of 0.499, and ROC-AUC of 0.777. The model had a high value for recall which indicated it was successful in detecting customers who are most likely to churn. However, the low precision of the model suggested that there were a lot of false-positives or incorrect classifications. Thus, while the model did have good success in predicting who would churn, it had poor precision. This could cause some issues in how you allocate your resources to retain customers.

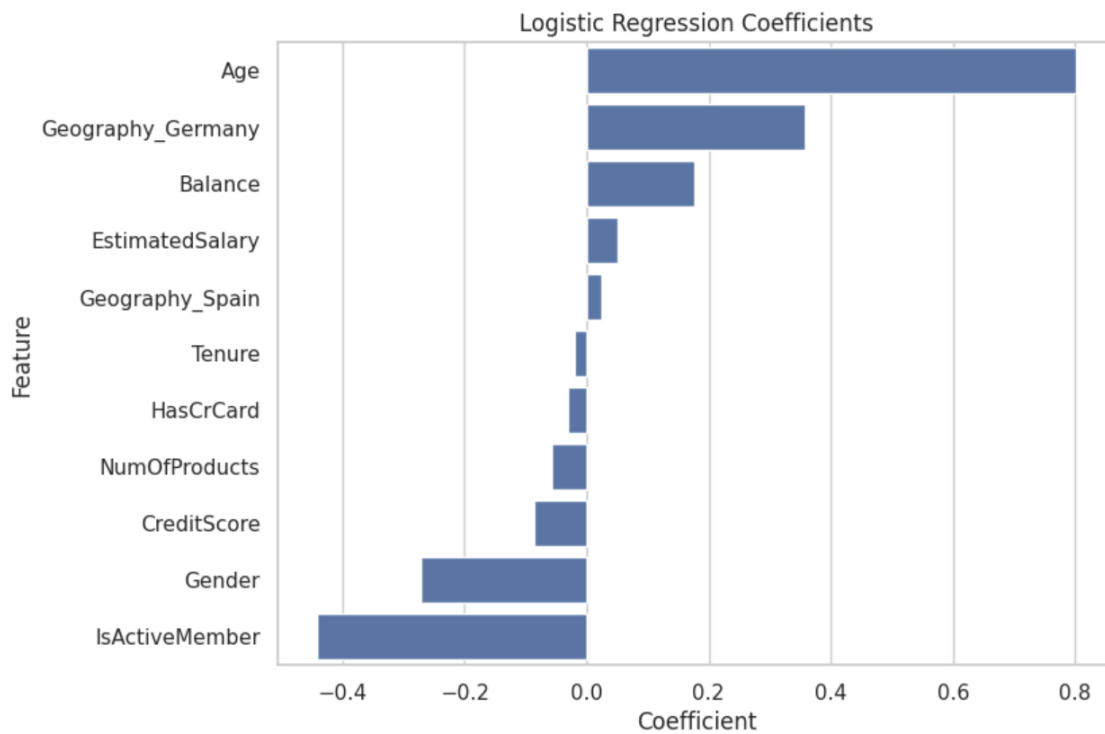
5.1.3 Confusion Matrix Interpretation

The confusion matrix showed that the model predicted 1,142 of 1,427 (or about 79.8%) retained customers and 285 out of 317 (or about 90.0%) churned customers. However, the model made 451 false positives (retained when they should be churned) and 122 false negatives (churned when they should be retained). Overall, these results show that the model was able to predict many churned customers, but still had a large amount of misclassification, especially in terms of retaining customers.



5.1.4 Coefficient Interpretation

An examination of the coefficients from the logistic regression found Age (+0.802) to be the largest positive predictor of churn. This was followed closely by Geography_Germany (+0.358) and then Balance (+0.176), showing that older customers, German customers and those with larger balances are more likely to churn. On the other hand, both IsActiveMember (-0.442) and Gender (-0.272) showed inverse relationships with churn suggesting that being an active member and having gender characteristics associated with non-churners will result in a decreased likelihood of leaving the bank.



	Feature	Coefficient
2	Age	0.801883
9	Geography_Germany	0.357669
4	Balance	0.175751
8	EstimatedSalary	0.050502
10	Geography_Spain	0.023974
3	Tenure	-0.018831
6	HasCrCard	-0.031611
5	NumOfProducts	-0.057734
0	CreditScore	-0.087119
1	Gender	-0.272170
7	IsActiveMember	-0.442434

5.1.5 Section Conclusion

Overall, Logistic Regression was very interpretable and helped provide a lot of insight on what drove customers' behavior. It is therefore beneficial in terms of providing an overall understanding of how managers can make decisions regarding their business. Nevertheless, while the logistic regression model identified many of the most important predictor variables, it did not perform well at all when compared to other machine learning models like Random Forest and Gradient Boosting in predicting churn.

5.2 Random Forest Results

5.2.1 Model Overview

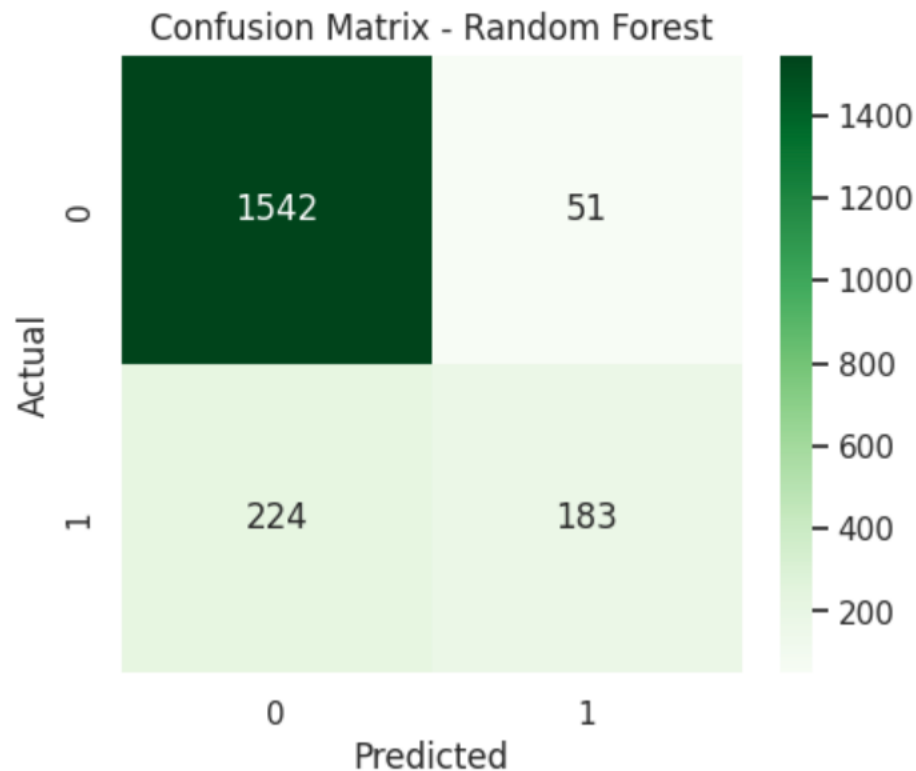
Random forest was used to create a better predictive model using ensemble learning techniques and creating multiple decision trees. This method was chosen due to its high degree of robustness, its ability to identify complex, nonlinear relationships, and its ability to determine feature importance. It is ideal for modeling customer churn (Breiman, 2001) .

5.2.2 Model Performance

The accuracy of the random forest model is approximately 86.2% , the precision of the random forest model is approximately 78.2% , the recall of the random forest model is approximately 45.0 % , the F1 score of the random forest model is approximately .571 , and the roc-auc of the random forest model is approximately 0.855. As compared to the Logistic Regression model, the random forest model has much better predictive performance. Also, the precision of the random forest model is greater than that of the Logistic Regression model. The lower recall indicates that the random forest model did not detect all at risk customers.

5.2.3 Confusion Matrix

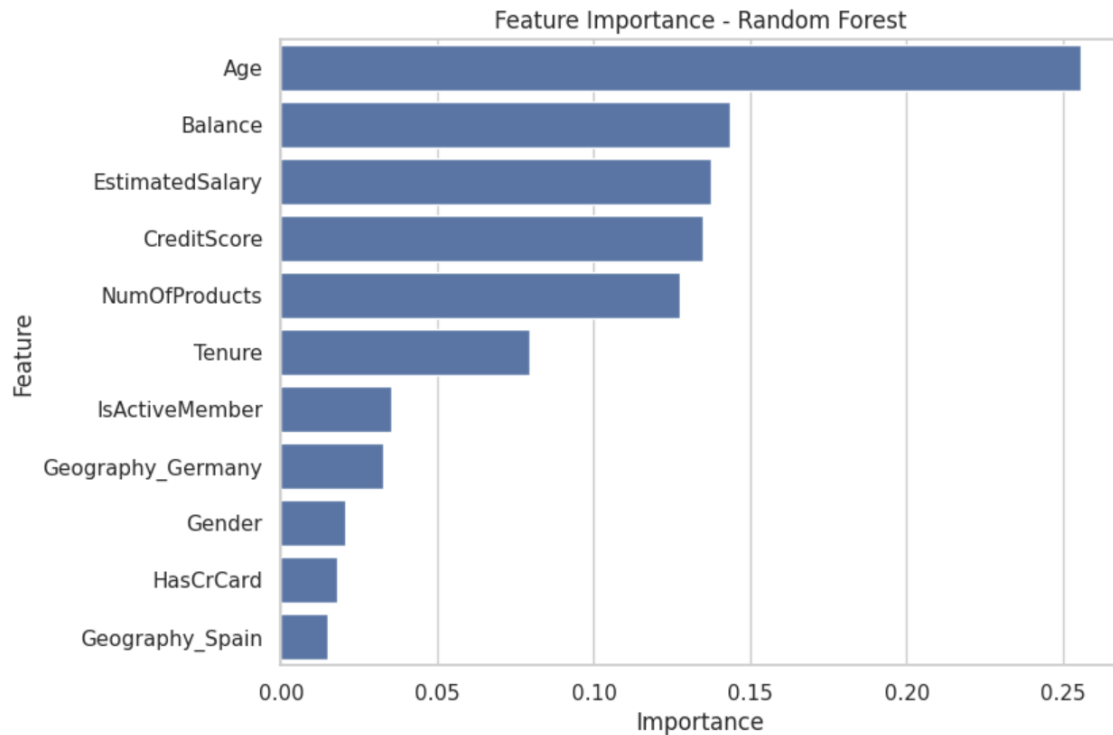
The confusion matrix shows that the model accurately predicted 1,542 retained customers and 183 churned customers. In addition, there were 51 false positives and 224 false negatives. Therefore, the random forest model generated less false positive churn predictions than the Logistic Regression model, however, it failed to predict a larger number of actual churned customers.



	precision	recall	f1-score	support
0	0.87	0.97	0.92	1593
1	0.78	0.45	0.57	407
accuracy			0.86	2000
macro avg	0.83	0.71	0.74	2000
weighted avg	0.85	0.86	0.85	2000

5.2.4 Feature Importance Analysis

The feature importance analysis determined that Age (0.256) is the most important variable in predicting whether or not a customer will churn from a company. The second most important factor in determining whether or not a customer will churn from a company are Balance (0.143), estimated salary (0.137), credit score (0.135) and number of products purchased (0.127). All of these variables have been previously identified as significant factors in customer retention based on previous data exploration activities. The consistent Results between both exploratory and predictive models increase my confidence in identifying the most critical factors leading to customer churn.



	Feature	Importance
2	Age	0.255595
4	Balance	0.143344
8	EstimatedSalary	0.137498
0	CreditScore	0.134865
5	NumOfProducts	0.127452
3	Tenure	0.079522
7	IsActiveMember	0.035532
9	Geography_Germany	0.032779
1	Gender	0.020483
6	HasCrCard	0.017883
10	Geography_Spain	0.015046

5.2.5 Section Conclusion

In summary, the use of the random forest model significantly increased the accuracy and strength of classification when compared to Logistic Regression. Although it provided a strong classification performance, it also resulted in a lower recall value indicating that it may not be able to provide complete identification of churned customers.

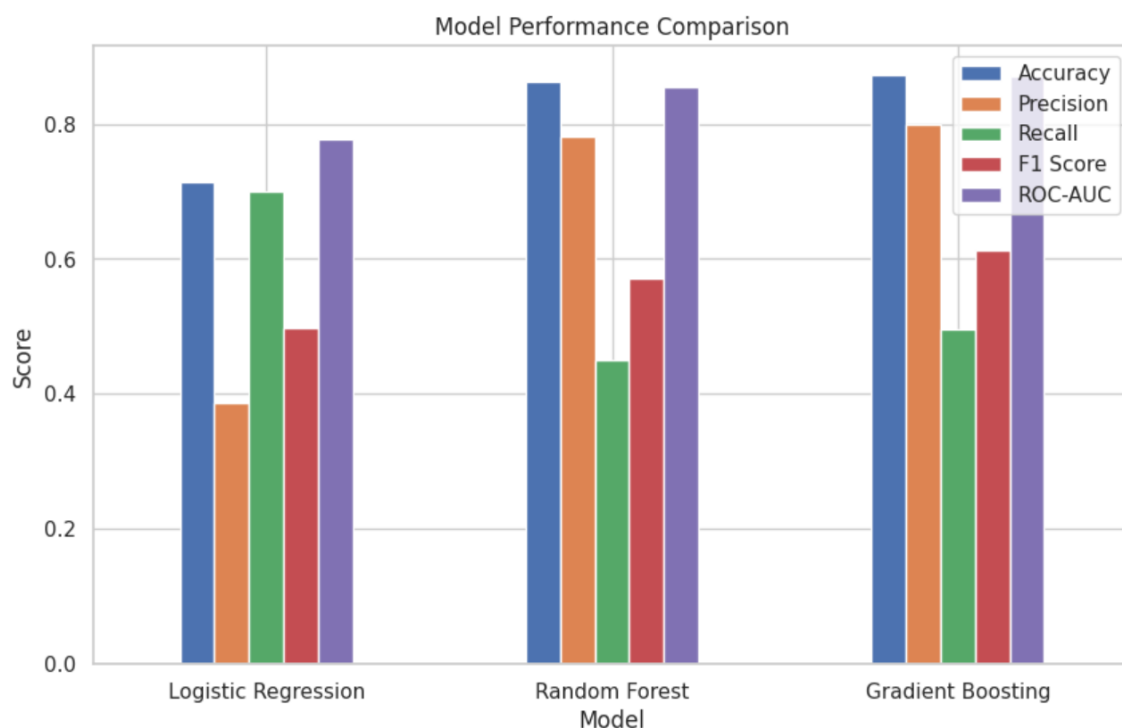
5.3 Gradient Boosting Results

5.3.1 Model Overview

Gradient Boosting was used because it uses a sequential approach to improve predictive accuracy by building one decision tree on top of another. Each new tree tries to correct for mistakes in prior trees. This is suitable for this project since Friedman (2001) indicates that Gradient Boosting has strong classification capabilities and can identify complex customer behavior.

5.3.2 Model Performance

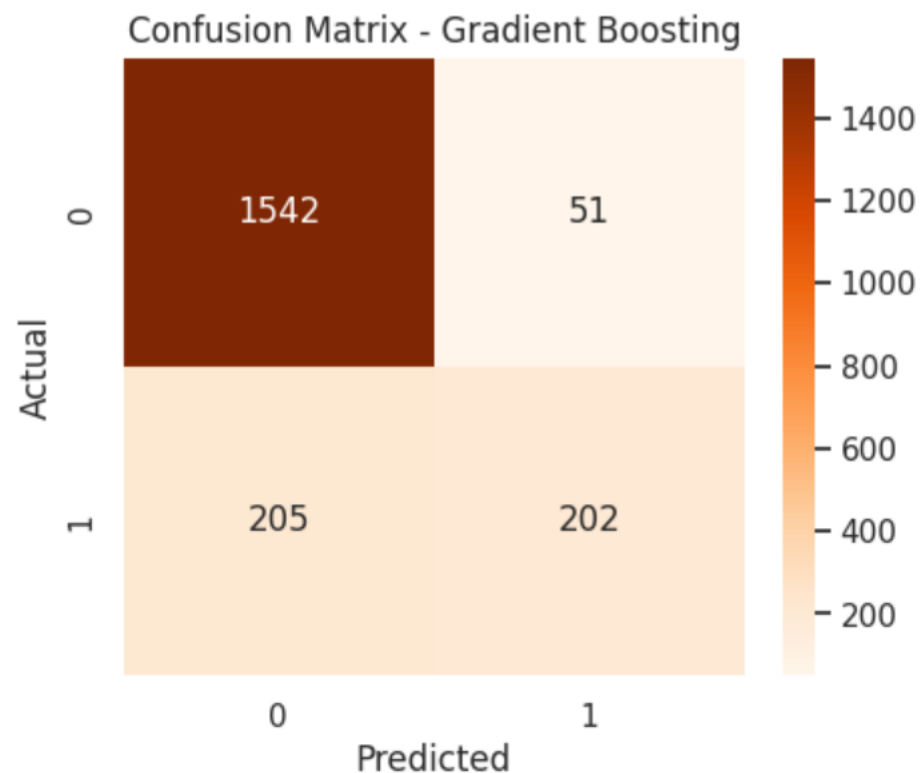
The Gradient Boosting model performed very well when using metrics for evaluation. The model had a classification accuracy of 87.2%, a precision of 79.8%, a recall of 49.6%, an F1-Score of 0.612, and a ROC-AUC of 0.871. When comparing the three models used in this research, Gradient Boosting outperformed both the other models based on highest accuracy value, the highest F1-Score value, and the highest ROC-AUC value. In comparison to the previous two models, the Gradient Boosting model shows a better balance of true positive identification and false negative/positive reductions.



5.3.3 Confusion Matrix

In addition to providing some insight into how many customers have been accurately predicted as retained or churned, the confusion matrix also provides insight into how many false positives and false negatives were generated by the model. Based upon the data generated, the Gradient Boosting model was able to predict that 1,542 retained customers were retained and 202 churned customers were actually churned. At the same time however, there were 51 false positives and 205 false negatives. As shown in Table 4-7, the Gradient

Boosting model did a much better job than either of the previous models at predicting which customers would churn.



	precision	recall	f1-score	support
0	0.88	0.97	0.92	1593
1	0.80	0.50	0.61	407
accuracy			0.87	2000
macro avg	0.84	0.73	0.77	2000
weighted avg	0.87	0.87	0.86	2000

5.3.4 Section Conclusion

As indicated above, the Gradient Boosting model was the most accurate predictive model used in this research study.

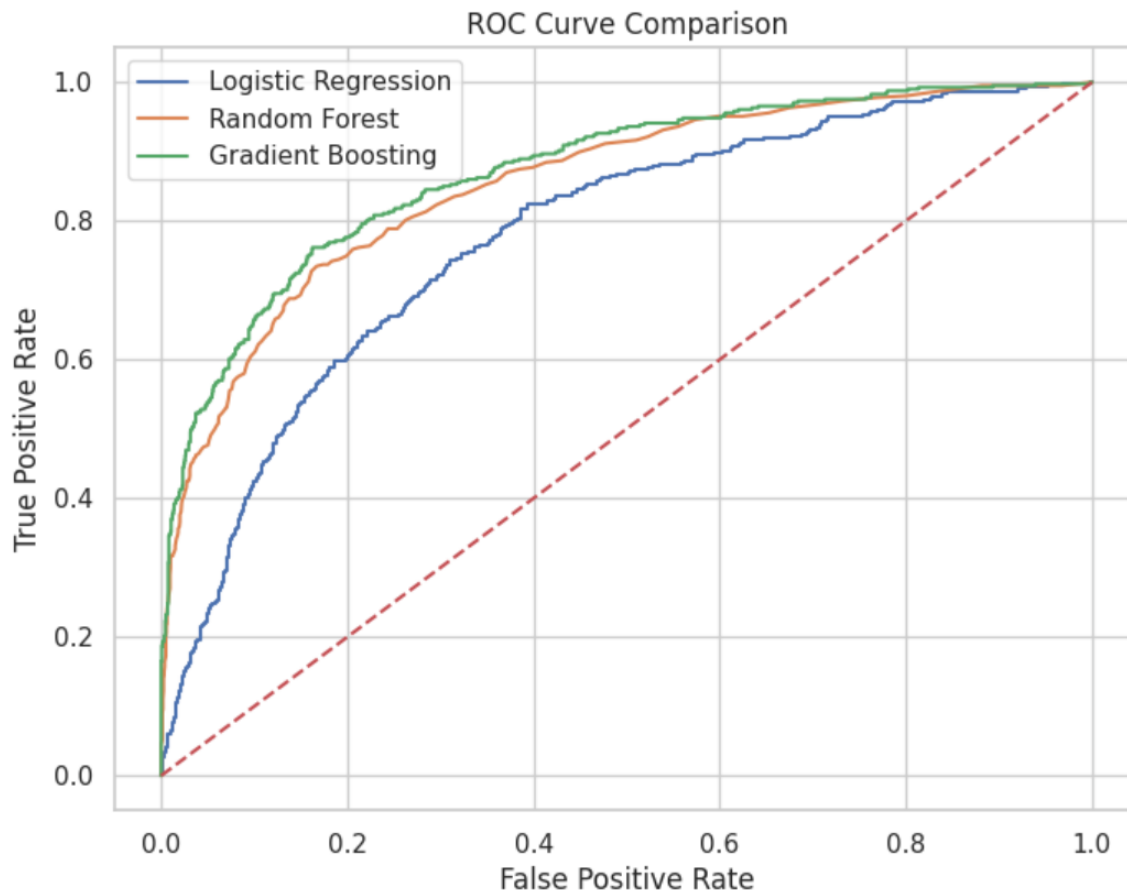
5.4 Cross-Validation Results

Cross validation is used as an additional method to evaluate the reliability of the model's predictions. A single training/test split may be a poor indicator of how well a model will perform in other scenarios. In this case, the average ROC-AUC scores from multiple cross

validations were calculated. Gradient boosting had the highest average ROC-AUC at 0.864, followed closely by Random forest at 0.858, and the lowest performance was recorded for Logistic regression at 0.767. All of the three models also exhibited very low standard deviations of .010 to .012, which indicates very good consistency or stability among their predictions using different data splits. This suggests that the models are reliable and not too much dependent upon how the data is divided. Therefore, the best performing model was gradient boosting, which also demonstrated the best robustness and predictive power. Thus, it can be concluded that gradient boosting is most suitable for predicting customer churn in the bank environment.

5.5 ROC Curve Comparison

ROC Curve comparison is used to assess how well each Machine Learning Model can classify churned vs. non-churned Customers. By using the ROC-AUC Values, Gradient Boosting has the best Performance of all Models at 0.871. This is followed by Random Forest with a value of 0.855 and lastly, Logistic Regression with a Value of 0.777. The ROC Curve for Gradient Boosting lies closest to the upper left corner which indicates a high degree of discriminative power and strong Classification Power. On the other hand, Logistic Regression shows significantly lower discrimination. Thus these results provide evidence that Gradient Boosting is preferable for use as the predictive model for banking Customer Churn Prediction.



5.6 Final Model Comparison

Comparison of Machine Learning models showed a significant difference in ability to predict and in their ability to be interpreted. logistic regression is easy to understand by managers because it has good explanatory power and transparency but had less predictive power than Ensemble models. Random Forest performed well on Accuracy and Precision which reduced False Positives and had better Recall than Logistic Regression to identify All Churned Customers. On the other hand, Gradient Boosting performed best in Overall with the highest Accuracy (87.2 %), F1-Score (0.612), and ROC-AUC Value (0.871) of the three models. Additionally, it had better Balance between Precision and Recall that helped Gradient Boosting to Identify Customer Churn. Therefore, Gradient Boosting was selected as the Preferred Predictive Model Because of Higher Classification.

CHAPTER 6 — BUSINESS INSIGHTS & DISCUSSION

6.0 Introduction

Exploratory Data Analysis & Machine Learning Models in Customer Churn Prediction, Business Interpretation, Managerial Implications of Churn, Key Drivers of Churn, Practical Retention Strategies to Enhance Long-Term Customer Relationships & Profitability." The understanding of churn behavior in banking is especially relevant due to the significance that customer retention has in contributing to competitive advantages as well as sustainable business growth (Kotler and Keller, 2016)

6.1 Key Churn Drivers

Several important influencers of customer churn have been identified based on the findings of Chapter 4 and Chapter 5. The variable that has consistently emerged as the most influential factor is age. For example, age was shown to be the most consistent influencer of churn in each of the three analyses used in this research project: correlation, logistic regression (LR) coefficient analysis, and random forest feature importance analysis. In each case, older customers were significantly more likely than younger customers to churn from their bank. Possible reasons for this include: increasing priority placed upon financial needs during later years of life, greater expectations regarding service quality in later years of life, and/or dissatisfaction with their current bank's product offerings and/or banking experience. In terms of geographic location, Germany had the greatest number of customers who churned from their banks, which suggests that there may exist some geographic influences on customer behavior. These possible geographic influences could be related to differences in: customer satisfaction with their local banks, level of competition in local markets, or different customer expectations related to the delivery of banking services in regional areas (Verbeke et al., 2011).

Finally, the results indicate that a positive relationship exists between customer inactivity and churn. As such, disengagement appears to lead to a decrease in loyalty toward a bank. Therefore, it can be argued that customer engagement is critical to achieving long-term retention (Grönroos, 1994).

6.2 Business Impact of Churn

The customer churn can result in severe financial issues to the banking industry, because the costs related to attracting a new client are much higher than maintaining an existing one (Kotler & Keller, 2015). This research has shown that about 20% of the bank's clients have been leaving. This is a major threat to long term viability and sustainability of the company as well as its profitability. The customer churn can lead to reduction in recurring income from financial products/services as well as to decrease in Customer Lifetime Value (CLV). In addition, there will be increase in expenses to replace the lost customers through advertising, promotions and onboarding. Therefore, predictive analytics enables the banks to detect the risk to lose their customers and apply appropriate preventive measures to minimize losses.

6.3 Retention Recommendations

1. Banks should develop age-based customer segmentation (e.g. Age 60+) and offer customised products and services to help long term engagement of these ages.
2. Also a Germany focused retention strategy must be developed as there were higher churn levels in this region. This may be due to dissatisfaction with service quality or lack of location specific marketing initiatives.
3. campaigning for inactive account holders should be prioritized to enhance activity by low usage customers. Personalised loyalty programmes, communication and offers may encourage continued banking activity and reduce switching tendencies.
4. Finally banks should strengthen their high balance protection strategies, as both high value and those who have the potential to turn over money also demonstrated churn. Premium banking services and relationship management may improve retention.

Risk Group	Recommendation
Older Customers	Personalized products
Germany Customers	Regional Strategy
Inactive Customers	Re-engagement campaigns
High Balance Customers	Premium retention

6.4 Performance vs Explainability

Performance of Models is Tied to the Interpretability of Those Same Models.

Interpretability was found to be in direct conflict with the predictive power of models.

Logistic Regression, provided an interpretable method of prediction because coefficients are directly observable and this makes logistic regression well suited as a tool that managers can use for explanations or where regulatory considerations exist. Conversely, gradient boosting jamealso limits how much interpretive analysis you can do with gradient boosted models. random forest represents a balance point for both predictive ability and interpretative capabilities because random forests provide better predictive results than logistic regression, however, they offer some measure of interpretability through feature importance. As such, banks will have options based on their business needs when selecting among these methods of predicting customer churn (Molnar, 2022).

6.5 Chapter Summary

The primary factors that drive customer churn were presented throughout this chapter, these include age, location (geography), lack of usage/inactivity, and an unbalanced account, along with a discussion of the business implications associated with each. Also, the Gradient Boosting model was identified as the most effective method for building a predictive model. This is followed by the next chapter's overall conclusions, the potential limitations to this research project, and recommendations for future studies on this topic.

CHAPTER 7 — CONCLUSION

7.0 Chapter Introduction

The purpose of this chapter is to present a summary of the conclusions of this study. This will include an overview of the key findings from this study, research contributions that have resulted from this study, limitations and potential avenues for further research based on the outcomes of this study.

7.1 Summary of Key Findings

This research shows there is a significant issue with retaining customers in the banking sector due to a high churn rate of 20.37%. From the EDA, it was also shown that female customers had the highest churn rates and Germany has the largest percentage of churned customers. Older customers, inactive customers and customers with larger account balances are more likely to churn. Using a predictive modeling approach, logistic regression provided good interpretability, however, the ability to classify was limited compared to random forest. The results from gradient boosting showed superior classification capability compared to both logistic regression and random forest. The model achieved an overall accuracy of 87.2%, F1 score of 0.612, and ROC-AUC of 0.871. Therefore, gradient boosting has the greatest ability to predict whether or not a customer will churn out of a banking dataset (Friedman, 2001, James et al., 2021)

7.2 Contribution of the Study

The research added to customer churn predictions studies using three model types that compare logistic regression with random forest and gradient boosting models within an industry bank. In addition, this research combines predictive analysis with managerial interpretation for practical application. This research adds to current churn prediction

research as it evaluates both predictive modeling results with managerial interpretation. (Verbeke et al., 2011)

7.3 Limitations of the Study

- 1) The research results are based upon one dataset from Kaggle (which includes 10,000 bank customers), therefore it is possible that these may not reflect all types of banks or their customers.
- 2) There were no actual banking customer data included in the features used for the analysis. This could potentially limit how broadly applicable the results might be.

7.4 Recommendations for Future Research

In order to develop a better model of predicting bank customer loss using data, we recommend that researchers collect large amounts of data from real banks in other countries. We also recommend that researchers develop models based on deep neural networks, develop methods of artificial intelligence (xai) to provide explanation of results, and make an analysis of the differences between banks in different countries.

7.5 Final Concluding Statement

The use of Machine Learning Techniques will be a major asset in Proactive Customer Retention Strategies to build on Customer Relationship Management (CRM) that will have an impact on Long-Term Profitability for Banks. As a result of effective customer churn predictions via machine learning, banks will be able to develop their ability to retain customers as well as make informed decisions which will ultimately lead to longer term sustainable business operations.

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APPENDICES

Appendix A — Dataset Description

Variable	Description	Type
CreditScore	Customer Credit Score	Numerical
Geography	Customer Country	Categorical
Gender	Male/Female	Categorical
Age	Customer age	Numerical
Tenure	Years with bank	Numerical
Balance	Account balance	Numerical
NumOfProducts	Number of Products	Numerical
HasCrCard	Has Credit Card (0/1)	Binary
IsActiveMember	Active Member (0/1)	Binary
EstimatedSalary	Estimated Salary	Numerical
Exited	Churn status (0 = retained, 1 = churned)	Target variable

Source: Kaggle Bank Customer Churn Dataset (10,000 observations).
Kaggle (n.d.) *Bank Customer Churn Dataset*. Available at: <https://www.kaggle.com/>

Appendix B — Data Preprocessing Steps

B1. Missing Value Check

```
print("\nMissing Values")
print(primary_df.isnull().sum())
```

Result: No missing values were detected.

```

DATASET INFORMATION
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   RowNumber              10000 non-null  int64
1   CustomerId             10000 non-null  int64
2   Surname                10000 non-null  object
3   CreditScore            10000 non-null  int64
4   Geography              10000 non-null  object
5   Gender                 10000 non-null  object
6   Age                    10000 non-null  int64
7   Tenure                 10000 non-null  int64
8   Balance                10000 non-null  float64
9   NumOfProducts          10000 non-null  int64
10  HasCrCard              10000 non-null  int64
11  IsActiveMember         10000 non-null  int64
12  EstimatedSalary         10000 non-null  float64
13  Exited                  10000 non-null  int64
dtypes: float64(2), int64(9), object(3)
memory usage: 1.1+ MB

```

```

Missing Values
RowNumber      0
CustomerId     0
Surname        0
CreditScore    0
Geography      0
Gender         0
Age            0
Tenure         0
Balance        0
NumOfProducts  0
HasCrCard      0
IsActiveMember 0
EstimatedSalary 0
Exited         0
dtype: int64

```

```

Duplicate Records
0

```

Descriptive Statistics

	RowNumber	CustomerId	CreditScore	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
count	10000.00000	1.000000e+04	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.00000	10000.000000	10000.000000	10000.000000
mean	5000.50000	1.569094e+07	650.528800	38.921800	5.012800	76485.889288	1.530200	0.70550	0.515100	100090.239881	0.203700
std	2886.89568	7.193619e+04	96.653299	10.487806	2.892174	62397.405202	0.581654	0.45584	0.499797	57510.492818	0.402769
min	1.00000	1.556570e+07	350.000000	18.000000	0.000000	0.000000	1.000000	0.00000	0.000000	11.580000	0.000000
25%	2500.75000	1.562853e+07	584.000000	32.000000	3.000000	0.000000	1.000000	0.00000	0.000000	51002.110000	0.000000
50%	5000.50000	1.569074e+07	652.000000	37.000000	5.000000	97198.540000	1.000000	1.00000	1.000000	100193.915000	0.000000
75%	7500.25000	1.575323e+07	718.000000	44.000000	7.000000	127644.240000	2.000000	1.00000	1.000000	149388.247500	0.000000
max	10000.00000	1.581569e+07	850.000000	92.000000	10.000000	250898.090000	4.000000	1.00000	1.000000	199992.480000	1.000000

B2. Duplicate Check

```

print("\nDuplicate Records")
print(primary_df.duplicated().sum())

```

Result: No duplicate records were identified.

```

DATASET INFORMATION
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   RowNumber              10000 non-null  int64
1   CustomerId             10000 non-null  int64
2   Surname                 10000 non-null  object
3   CreditScore             10000 non-null  int64
4   Geography               10000 non-null  object
5   Gender                  10000 non-null  object
6   Age                     10000 non-null  int64
7   Tenure                  10000 non-null  int64
8   Balance                 10000 non-null  float64
9   NumOfProducts           10000 non-null  int64
10  HasCrCard               10000 non-null  int64
11  IsActiveMember           10000 non-null  int64
12  EstimatedSalary          10000 non-null  float64
13  Exited                   10000 non-null  int64
dtypes: float64(2), int64(9), object(3)
memory usage: 1.1+ MB

```

```

Missing Values
RowNumber      0
CustomerId     0
Surname        0
CreditScore    0
Geography      0
Gender         0
Age            0
Tenure         0
Balance        0
NumOfProducts  0
HasCrCard      0
IsActiveMember 0
EstimatedSalary 0
Exited         0
dtype: int64

```

```

Duplicate Records
0

```

Descriptive Statistics

	RowNumber	CustomerId	CreditScore	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
count	10000.00000	1.000000e+04	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.00000	10000.000000	10000.000000	10000.000000
mean	5000.50000	1.569094e+07	650.528800	38.921800	5.012800	76485.889288	1.530200	0.70550	0.515100	100090.239881	0.203700
std	2886.89568	7.193619e+04	96.653299	10.487806	2.892174	62397.405202	0.581654	0.45584	0.499797	57510.492818	0.402769
min	1.00000	1.556570e+07	350.000000	18.000000	0.000000	0.000000	1.000000	0.00000	0.000000	11.580000	0.000000
25%	2500.75000	1.562853e+07	584.000000	32.000000	3.000000	0.000000	1.000000	0.00000	0.000000	51002.110000	0.000000
50%	5000.50000	1.569074e+07	652.000000	37.000000	5.000000	97198.540000	1.000000	1.00000	1.000000	100193.915000	0.000000
75%	7500.25000	1.575323e+07	718.000000	44.000000	7.000000	127644.240000	2.000000	1.00000	1.000000	149388.247500	0.000000
max	10000.00000	1.581569e+07	850.000000	92.000000	10.000000	250898.090000	4.000000	1.00000	1.000000	199992.480000	1.000000

B3. Feature Selection

```

primary_df = primary_df.drop(
    columns=['RowNumber', 'CustomerId', 'Surname']
)

```

The variables of Row Number, Customer Id, and Surname were eliminated in the preprocessed data set due to their role as a unique identifiers in the data set rather than predictive characteristics. The variables are not behavioral or financial related to predicting customer churn. Eliminating these identifier based variables can improve model performance by limiting what the model is considering when making predictions (James et al., 2021)

	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
3	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0

B4. Encoding

```
# Encode Geography using one-hot encoding
primary_df = pd.get_dummies(
    primary_df,
    columns=['Geography'],
    drop_first=True
)

# Encode Gender
primary_df['Gender'] = primary_df['Gender'].map({
    'Female': 0,
    'Male': 1
})

# Convert dummy variables to integer
primary_df['Geography_Germany'] = (
    primary_df['Geography_Germany'].astype(int)
)

primary_df['Geography_Spain'] = (
    primary_df['Geography_Spain'].astype(int)
)

# Preview updated dataframe
primary_df.head()
```

The categories of categorical data were converted to a numeric format so that they could be used in machine learning algorithms. The algorithm does not accept categorical variables in their textual representation. One-hot encoded the 'Geography' variable to produce two binary variables, one for 'Geography_Germany' and another for 'Geography_Spain'. However, because there is no valid method to include all three variables ('France') due to the presence of multicollinearity, it was left out as the base case or reference category in order to eliminate this issue (dummy variable trap). Label encoded the 'Gender' variable by assigning a value of '0' for 'Female' and '1' for 'Male' (James et al., 2021).

	CreditScore	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited	Geography_Germany	Geography_Spain
0	619	0	42	2	0.00	1	1	1	101348.88	1	0	0
1	608	0	41	1	83807.86	1	0	1	112542.58	0	0	1
2	502	0	42	8	159660.80	3	1	0	113931.57	1	0	0
3	699	0	39	1	0.00	2	0	0	93826.63	0	0	0
4	850	0	43	2	125510.82	1	1	1	79084.10	0	0	1

B5. Feature Scaling

```
scaler = StandardScaler()
```



```
X_train_scaled = scaler.fit_transform(X_train)

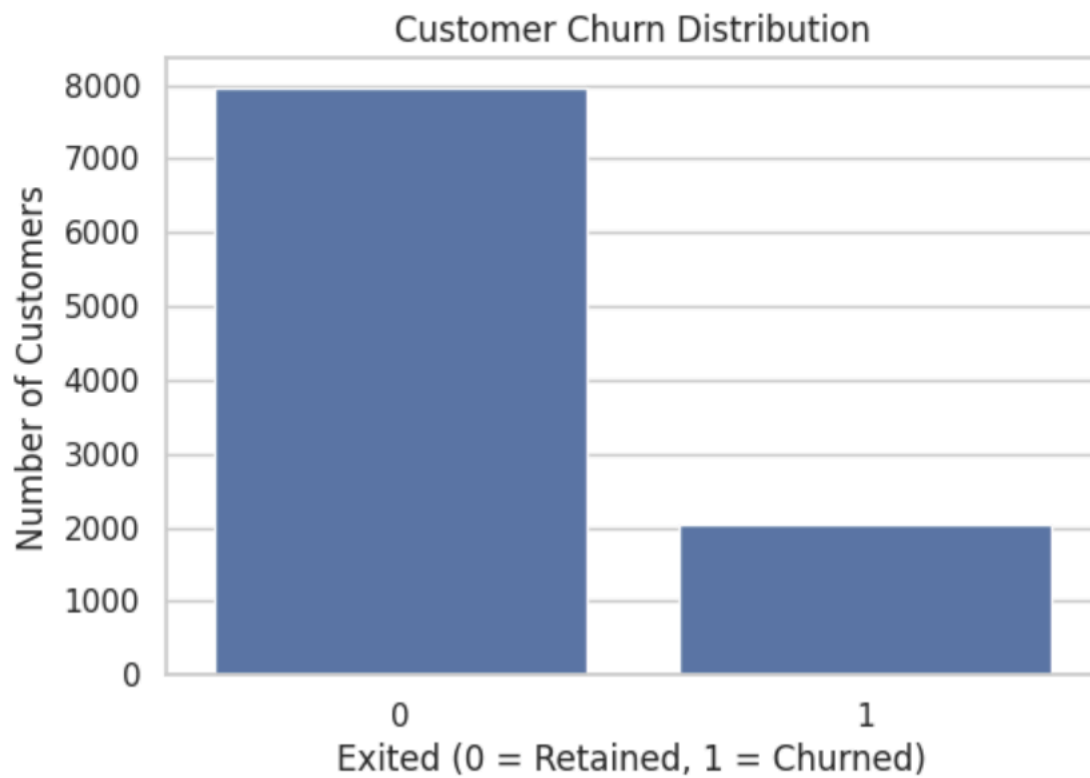
X_test_scaled = scaler.transform(X_test)

print("Feature scaling completed.")
```

Appendix C — Exploratory Data Analysis (EDA)

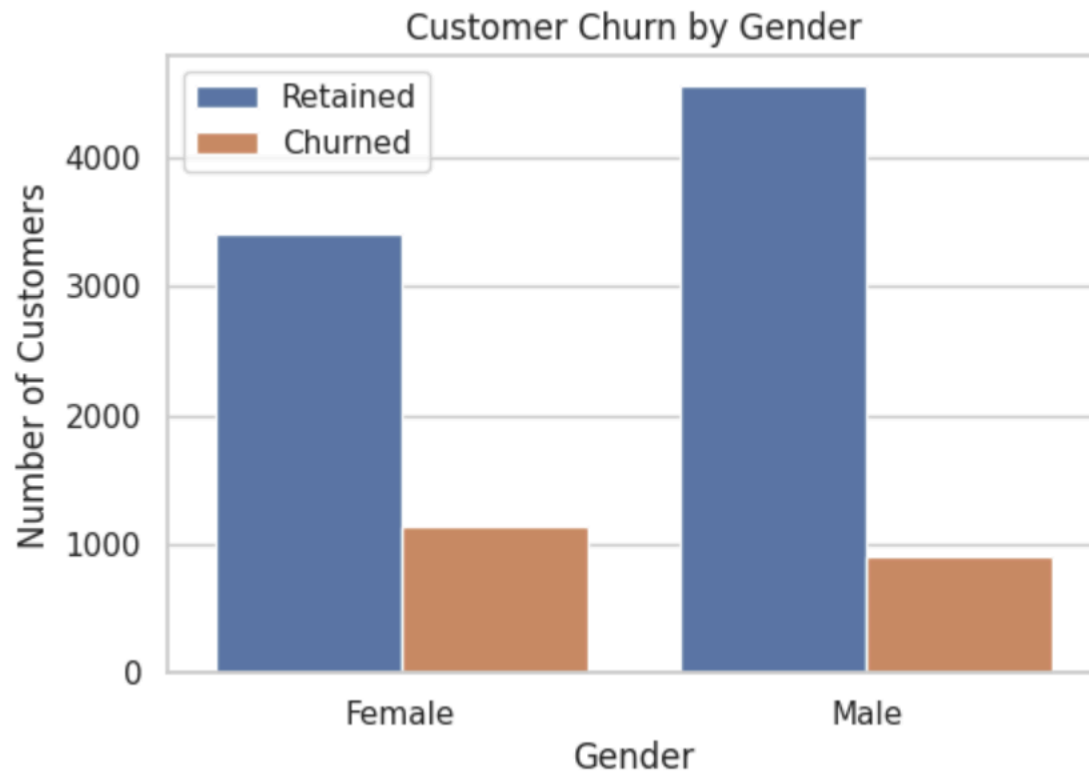
C1. Churn Distribution

Figure C1:



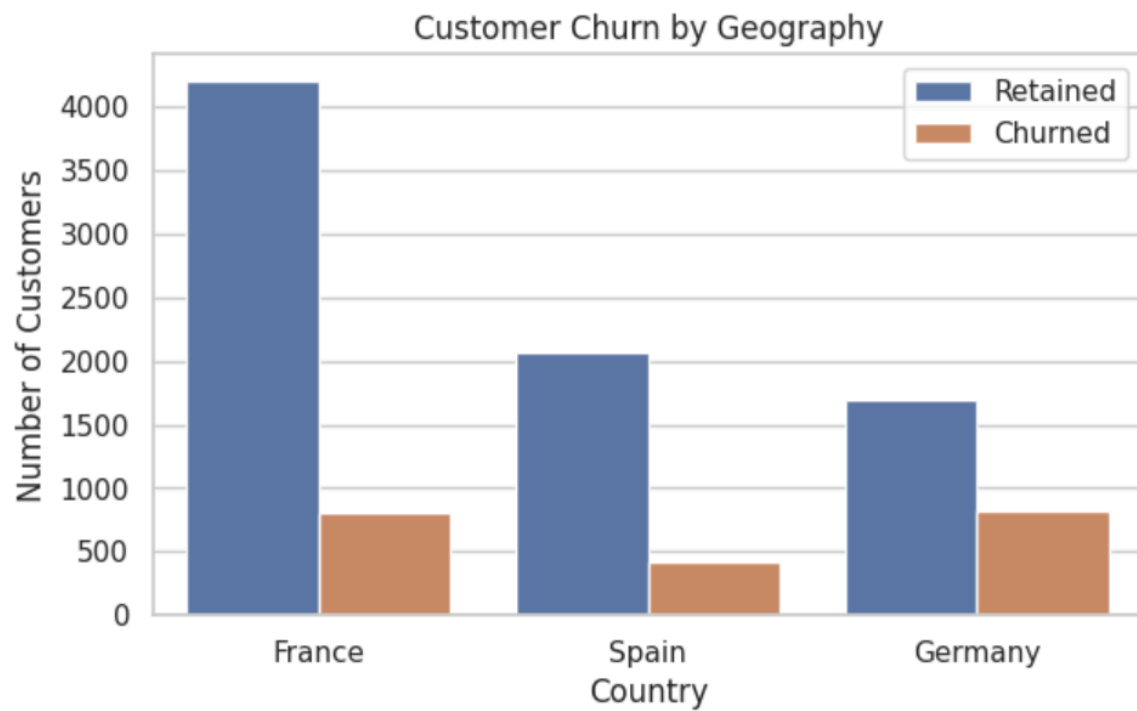
C2. Churn by Gender

Figure C2:



C3. Churn by Geography

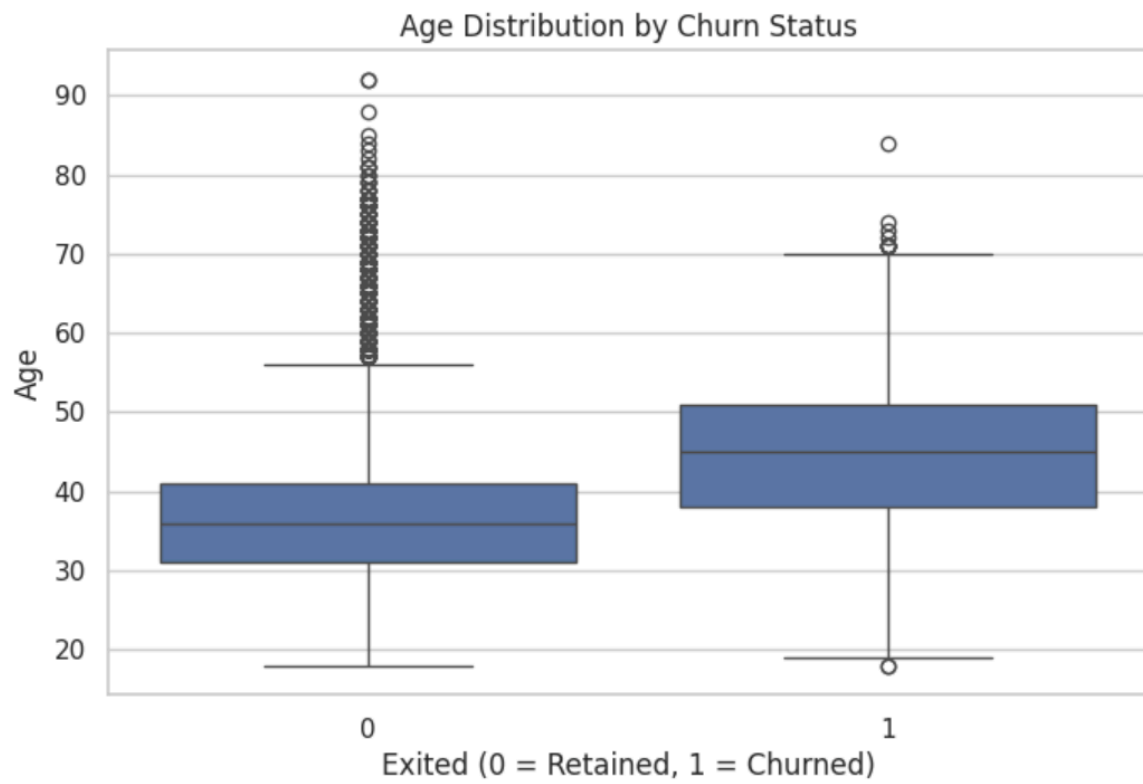
Figure C3:



Geography_Label	Exited	
	0	1
France	83.845233	16.154767
Germany	67.556796	32.443204
Spain	83.326605	16.673395

C4. Age and Churn Boxplot

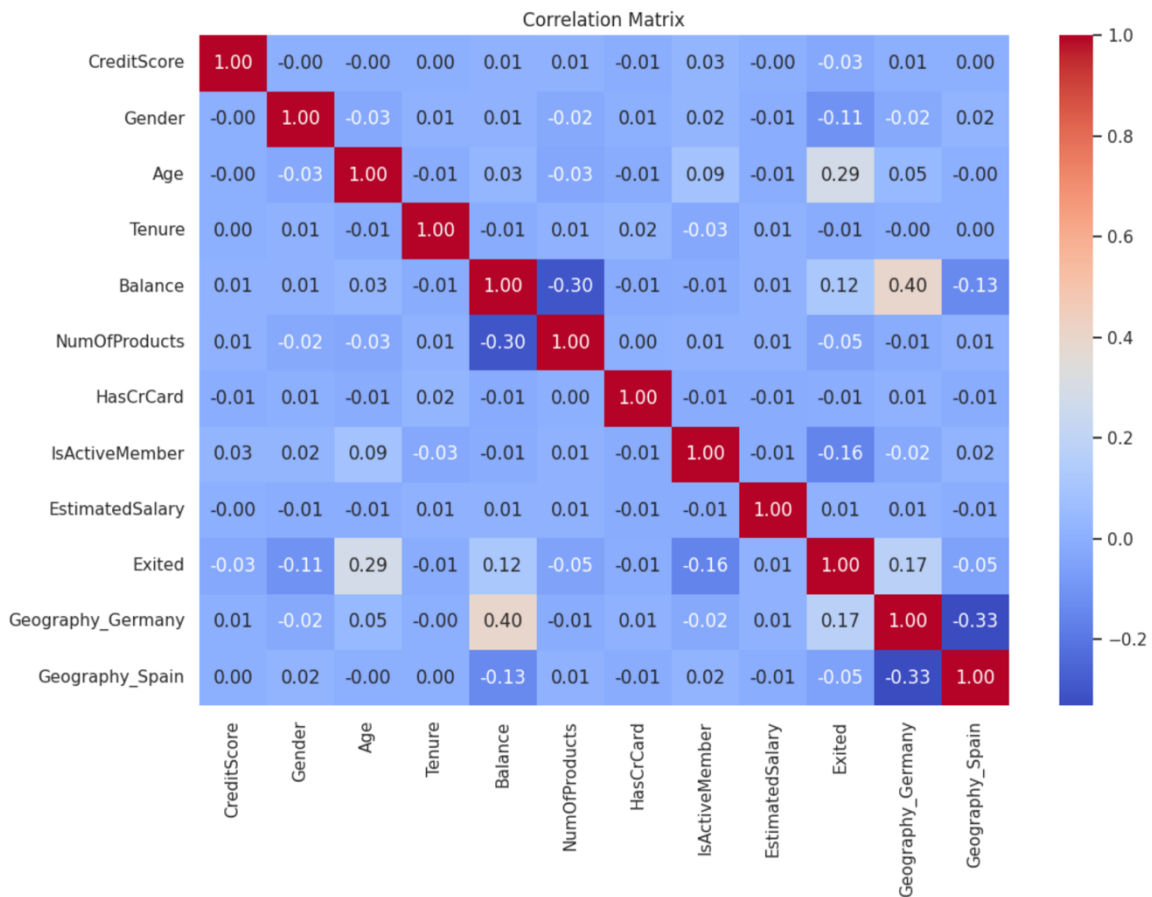
Figure C4:



	count	mean	std	min	25%	50%	75%	max
Exited								
0	7963.0	37.408389	10.125363	18.0	31.0	36.0	41.0	92.0
1	2037.0	44.837997	9.761562	18.0	38.0	45.0	51.0	84.0

C5. Correlation Heatmap

Figure C5:



Appendix D — Model Evaluation Results

D1. Model Performance Table

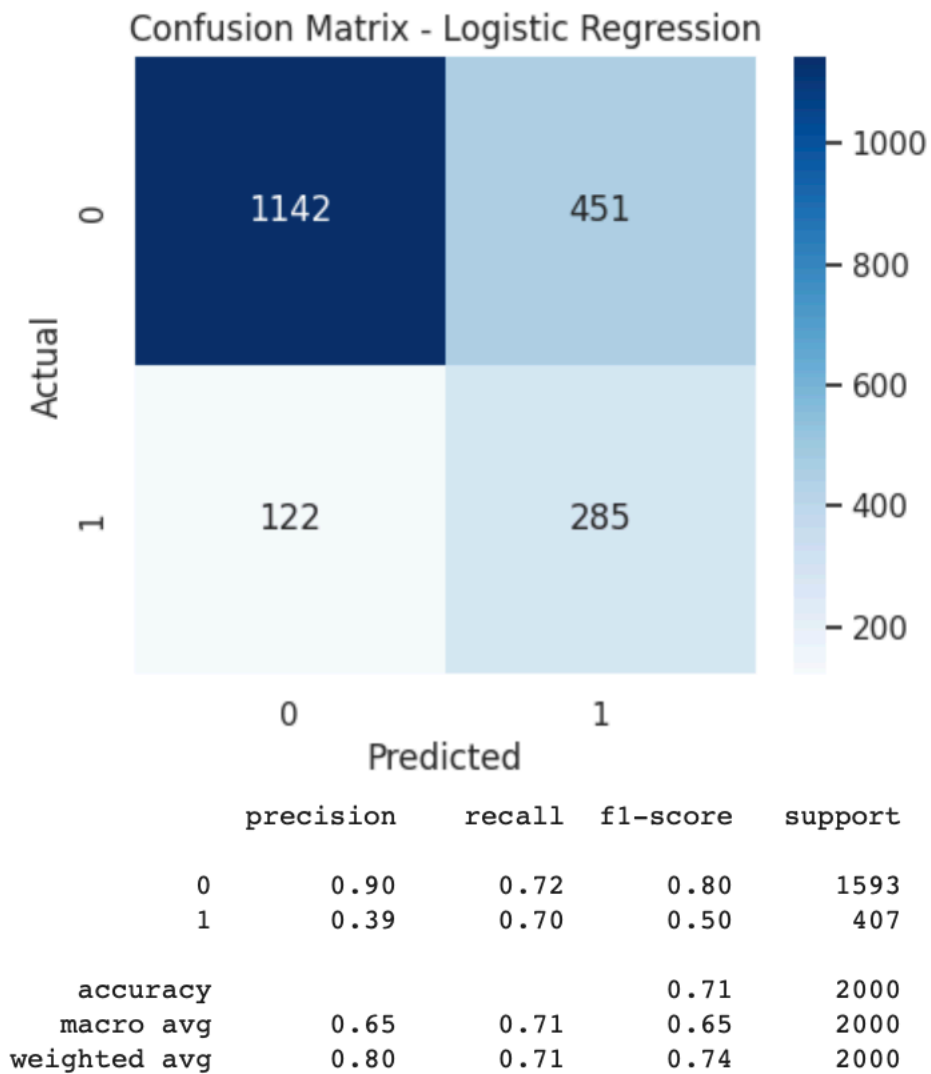
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.714	0.387	0.700	0.499	0.777
Random Forest	0.862	0.782	0.450	0.571	0.855
Gradient Boosting	0.872	0.798	0.496	0.612	0.871

D2. Cross-Validation Results

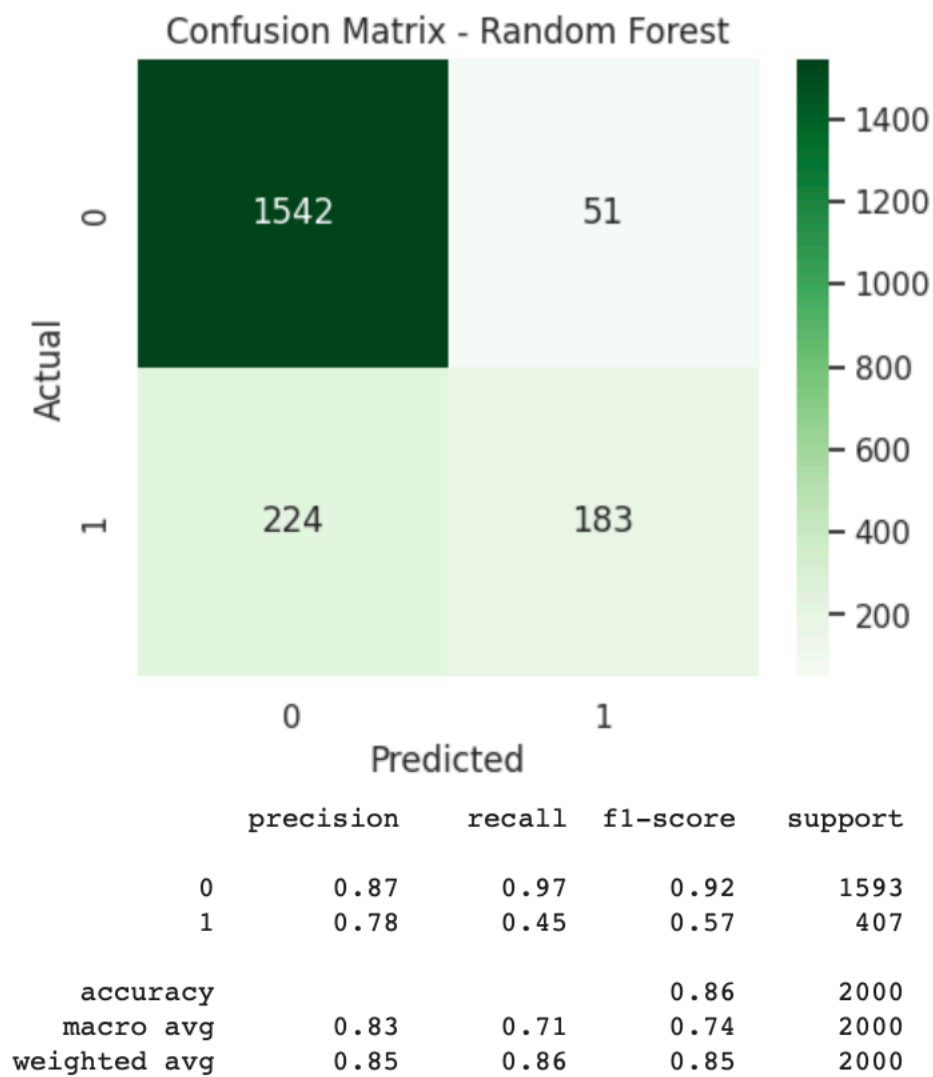
Model	Mean ROC-AUC	Standard Deviation
Logistic Regression	0.767	0.010
Random Forest	0.858	0.012
Gradient Boosting	0.864	0.011

Appendix E — Confusion Matrices

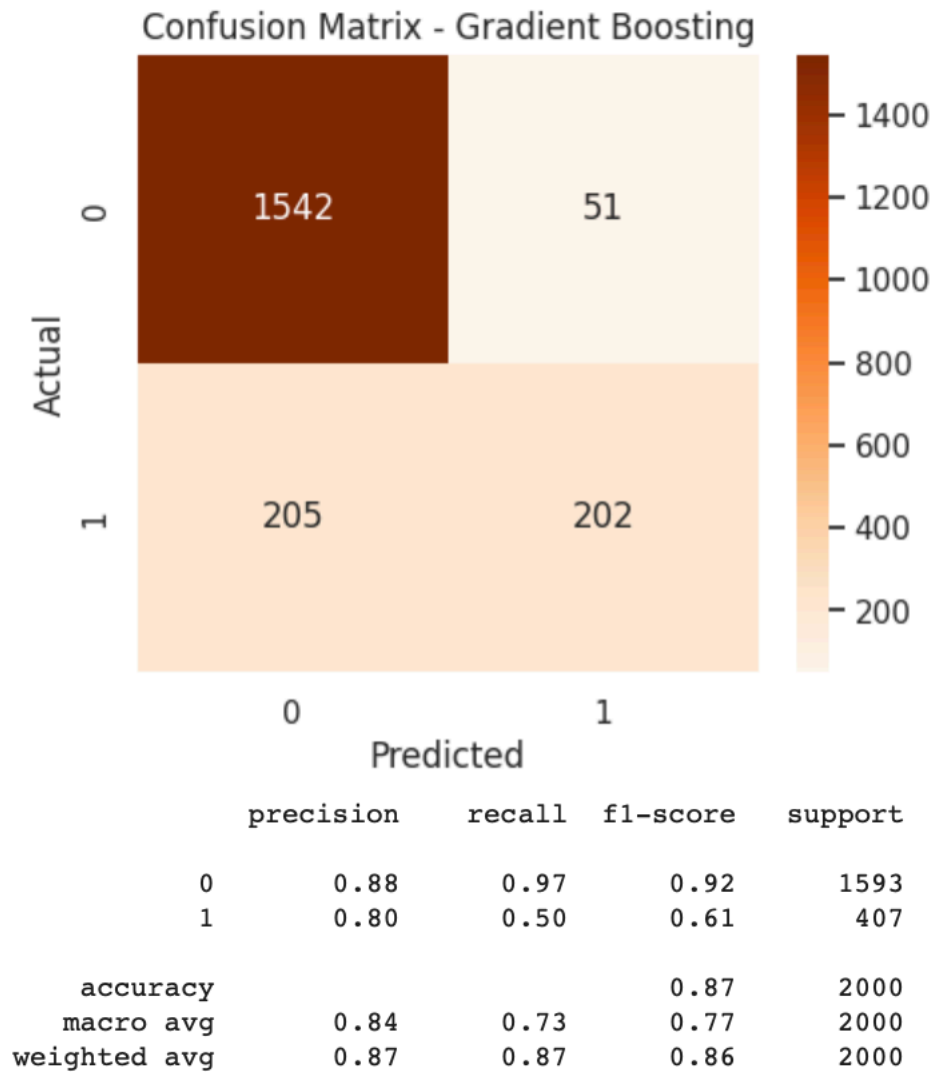
- **Figure E1:** Logistic Regression confusion matrix



• **Figure E2:**Random Forest confusion matrix



- **Figure E3: Gradient Boosting confusion matrix**



Appendix F — Python Code & Github Repository

GITHUB REPOSITORY AND PYTHON IMPLEMENTATION

All of the code implemented in this research project that was used for data cleaning and pre-processing, exploratory data analysis (EDA), feature engineering, implementing a Machine Learning Model, validating it, evaluating its performance are included in the python implementation that can be found here at the GitHub Repository below. Included with this is the full .ipynb notebook file that was used to complete all of the work on this business analytics project.

GitHub Repository Link:

<https://github.com/divyank3010/Bank-Churn-Analysis-Project/tree/main>

The repository includes:

- AI_LOG.md (Level 4 AI Documentation record.)
- Bank_Customer_Churn.csv (raw dataset and initial data cleaning scripts.)
- Business_Data_Analytics_Project.pdf (final business report and performance evaluation.)
- Divyank_Final-2.ipynb (mplement Logistic Regression, Random Forest, and Gradient Boosting models.)
- README.md
- Requirements.txt (Project structure and library dependencies.)

Appendix G — Ethics / AI Declaration (if required by uni)

AI tools helped generate ideas, structure ideas, provide proofread feedback on grammar and spelling errors and code suggestions for some python programming elements.

The researcher reviewed all data processing activities, the results from EDA, ML model development, interpreted the data results and made all final decisions about the results of this research without AI influence. The researcher evaluated all AI assisted output prior to using them in the final project.